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Review

Multi-UAV Task Assignment in Dynamic Environments: Current Trends and Future Directions

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Abstract: The rapid advancement of unmanned aerial vehicles (UAVs) has transformed a wide range of applications, including military operations, disaster response, agricultural monitoring, and infrastructure inspection. Deploying multiple UAVs to work collaboratively offers significant advantages in terms of enhanced coverage, redundancy, and operational efficiency. However, as UAV missions become more complex and operate in dynamic environments, the task assignment problem becomes increasingly challenging. Multi-UAV dynamic task assignment is critical for optimizing mission success. It involves allocating tasks to UAVs in real-time while adapting to unpredictable changes, such as sudden task appearances, UAV failures, and varying mission requirements. A key contribution of this article is that it provides a comprehensive study of state-of-the-art solutions for dynamic task assignment in multi-UAV systems from 2013 to 2024. It also introduces a comparative framework to evaluate algorithms based on metrics such as responsiveness, robustness, and scalability in handling real-world dynamic conditions. Our analysis reveals distinct strengths and limitations across three major approaches: market-based, intelligent optimization, and clustering-based solutions. Market-based solutions excel in distributed coordination and real-time adaptability, but face challenges with communication overhead. Intelligent optimization solutions, including evolutionary and swarm intelligence, provide high flexibility and performance in complex scenarios but require significant computational resources. Clustering-based solutions efficiently group and allocate tasks geographically, reducing overlap and improving efficiency, although they struggle with adaptability in dynamic environments. By identifying these strengths, limitations, and emerging trends, this article not only offers a detailed comparative analysis but also highlights critical research gaps. Specifically, it underscores the need for scalable algorithms that can efficiently handle larger UAV fleets, robust methods to adapt to sudden task changes and UAV failures, and multi-objective optimization frameworks to balance competing goals such as energy efficiency and task completion. These insights serve as a guide for future research and a valuable resource for developing resilient and efficient strategies for multi-UAV dynamic task assignment in complex environments.

Keywords: multi-UAV systems; dynamic task allocation; hybrid algorithm; UAV clustering techniques; real-time UAV coordination

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1. Introduction

The deployment of a group of robots in a cooperative manner to accomplish a set of tasks is a key topic of research in the robotics domain [1–3]. Such groups of robots, known

as multi-robot systems (MRSs), take advantage of the cooperative behaviors of robots to make feasible and attainable complicated tasks that are too time-consuming and exhausting for a single robot [2,4]. One of the most challenging problems in MRSs is the multi-robot task allocation (MRTA) problem. It concerns how to optimally assign a set of robots to a set of tasks in a way that optimizes predefined objectives and overall performance while considering a set of constraints [5].

The evolution and deployment of UAV technology in autonomous intelligent systems is one of the most crucial developments in the 21st century so far and has gained substantial attention in scientific research [6]. It has also significantly impacted a wide range of industries, including reconnaissance [7], disaster management [8], communication assistance [9], surveillance [10], and autonomous exploration [11]. Compared with a single-UAV system that has limited capabilities, multi-UAV systems, which involve the collaboration of multiple UAVs to complete intricate missions, have received widespread interest because of the advantages they offer, which include the following: collaboration and the distribution of workload among UAVs, the availability of different UAV capabilities, being more fault-tolerant to a single-point failure, and being more scalable in handling large problems [12,13]. Because of these advantages, multi-UAV systems are usually adopted to accomplish sophisticated tasks in complex environments. The multi-UAV task assignment problem is an instance of an MRTA problem that can be defined as a set of UAVs capable of performing a set of tasks in which the objective is to find an optimum assignment between UAVs and tasks while satisfying the requirements of predefined constraints [14–17]. The multi-UAV task assignment problem can be divided into two types according to the certainty of the environment: static or dynamic. Static task assignment assumes that the number of UAVs and tasks is fixed and that their status and location are known before allocation starts. In dynamic task assignment, the environment is constantly changing; new tasks may suddenly emerge during allocation, or one of the existing UAVs may fail to complete its mission [7]. The unpredictability of these environments requires algorithms that can rapidly adapt to changes, reassign tasks, and ensure mission success even when facing disruptions. This complexity is further compounded by the need to optimize resource allocation while managing communication constraints, ensuring system robustness, and maintaining the overall efficiency of the UAV fleet. This makes the problem of dynamic task assignment technically challenging and critical to the successful deployment of multi-UAV systems in real-world scenarios [18]. Such a problem is a combinatorial optimization problem known to be an NP-hard problem and obtaining a globally optimal solution becomes increasingly computationally extensive as the problem size increases [7,19].

This article tackles the state-of-the-art solutions to multi-UAV task assignment problems in dynamic environments in which uncertain events may occur at any time during execution. Over the past decade, researchers have proposed various solutions aimed at addressing these challenges, focusing on enhancing performance, robustness, and scalability across different operational contexts. We review studies published between 2013 and 2024 and examine the strengths and weaknesses of these solutions to identify prevailing trends, highlight research gaps, and propose directions for future studies in this rapidly evolving field. This article is organized as follows: Section 2 reviews previous surveys on the multi-UAV task assignment problem and highlights the strengths of this review. Section 3 introduces the problem of multi-UAV task assignment, detailing its challenges and importance, and presents a mathematical formulation of the problem. Sections 4 and 5 review the major categories of multi-UAV dynamic task assignment solutions—market-based, optimization-based, and clustering algorithms—used to address this problem and discuss solution characteristics, their performance, and their relevance to the dynamic task assignment. Finally, Section 6 concludes the review by summarizing the findings and suggesting potential areas for future research.

2. Comparative Analysis with Previous Surveys on Multi-Robot and Multi-UAV Task Assignment

Several previous surveys and reviews have explored the task assignment problem within multi-robot and multi-UAV systems, providing valuable insights into various algorithms and methodologies. Table 1 summarizes the scope, strengths, limitations, and how this work adds value relative to these surveys.

Table 1. Comparison of previous surveys and contributions of this work.

Survey	Scope	Strengths	Limitations	How This Work Adds Value
Song et al. [20]	Mission planning, covering task assignment and path planning for multi-UAV systems.	Broad overview of the task assignment methods (e.g., mathematical programming, neural networks, heuristic algorithms).	Limited focus on dynamic environments; insufficient strategies for scalability and communication constraints.	Focuses specifically on dynamic environments, addressing adaptability, scalability, and robust communication in real-time.
Nunes et al. [21]	Taxonomy of multi-robot task allocation (MRTA) with a focus on temporal and ordering constraints.	Comprehensive taxonomy for MRTA problems based on the robot and task characteristics.	Lacks UAV-specific focus; does not explore dynamic task reallocation, UAV constraints, or communication limitations.	Provides UAV-specific insights into scalability, adaptability, and dynamic reallocation in multi-UAV systems.
Peng et al. [22]	Dynamic task allocation for UAV swarms targeting ground operations.	Detailed discussion of market-based and intelligent optimization algorithms.	Insufficient analysis of scalability, communication constraints, and algorithm performance under diverse dynamic conditions.	Includes an in-depth comparison of algorithms and their performance in dynamic and adversarial environments.
Seenu et al. [23]	Dynamic task allocation strategies in multiple-robot systems.	Categorizes strategies into market-based, optimization-based, behavior-based, and clustered.	Limited analysis of hybrid approaches, scalability, and performance under different environmental uncertainties.	Explores hybrid approaches, multi-objective optimization, and algorithm robustness in dynamic environments.
Poudel et al. [24]	Task assignment strategies for UAV networks in dynamic environments.	Focus on real-time decision-making and categorization of centralized, distributed, bio-inspired, and multi-fusion algorithms.	Limited performance comparison; underexplored emerging algorithms and UAV-specific constraints.	Provides a comprehensive comparative analysis of algorithms, emerging trends, and UAV-specific operational constraints.

Strengths of this review article compared to previous works: This article provides a more focused and comprehensive analysis of the multi-UAV task assignment problem specifically in dynamic environments, which distinguishes it from previous surveys in several key areas. Unlike earlier works, which have generally addressed multi-robot systems (MRSs) or multi-UAV systems without focusing specifically on dynamic task assignment, this review article targets the unique challenges posed by dynamic environments in which tasks and conditions can change unpredictably.

The detailed exploration of these dynamic settings is a critical contribution, as it delves into how various task assignment algorithms are designed to handle unexpected

events, such as the sudden appearance of new tasks, UAV failures, or changes in mission parameters—areas that have been under-explored in prior research. Moreover, it provides a thorough classification and analysis of solutions used for dynamic task assignments, including market-based, intelligent optimization-based, and clustering algorithms. Table 1 summarizes the comparison of this work with previous surveys, highlighting its contributions and strengths.

Another significant strength of this review article is its focus on dynamic task reallocation, highlighting how different algorithms manage the need for constant reassignment and re-prioritization of tasks after detecting dynamic events. This aspect is critical for maintaining operational robustness in dynamic environments and is often underrepresented in other studies. Furthermore, this work includes an extensive comparative analysis of algorithms under different dynamic conditions, offering valuable insights into their performance and trade-offs, thus helping researchers and practitioners make informed decisions when selecting or developing task assignment strategies.

This work also identifies emerging trends and research gaps, providing a forward-looking perspective that encourages innovation and guides future research efforts to address current limitations. It discusses multi-objective optimization in dynamic task assignments, balancing competing goals, such as minimizing energy consumption and maximizing task coverage—another area where previous surveys fall short. Finally, the discussion of the robustness and scalability of algorithms in dynamic settings provides crucial insights into their suitability for real-world applications, making this review a comprehensive resource for advancing research and practice in multi-UAV dynamic task assignment.

3. Problem Definition

The multi-UAV dynamic task assignment problem can be generally stated as follows: Given a set of N UAVs $\{U_1, U_2, U_3, \ldots, U_N\}$ with limited payload capabilities and a set of M tasks $\{T_1, T_2, T_3, \ldots, T_M\}$, where $M > N$, as well as changeable environments. The common objectives of the problem are to find an assignment of UAVs to tasks that optimize predefined objectives: maximizing the $Benefit$, which could be the number of performed tasks, UAV throughput, or task rewards, and minimizing the $Cost$, which could be the distance, waiting time, or energy. Furthermore, when a sudden task arises, the original assignment is dynamically adjusted, and the UAVs autonomously adapt to these changes. The mathematical formulation of the problem is as follows:

$$\begin{aligned} &\text{Maximize} \quad \sum_{i=1}^{N}\sum_{j=1}^{M} X_{i,j} \cdot \text{Benefit}(U_i, T_j) \\ &\text{AND} \\ &\text{Minimize} \quad \sum_{i=1}^{N}\sum_{j=1}^{M} X_{i,j} \cdot \text{Cost}(U_i, T_j) \end{aligned} \tag{1}$$

Subject to:

$$\begin{aligned} C_1: &\quad X_{i,j} \in \{0,1\}, \\ C_2: &\quad \sum_{j=1}^{M} X_{i,j} \le L_i, \quad \forall U_i \in U, \quad i \in [1,N], \quad j \in [1,M], \\ C_3: &\quad \sum_{j=1}^{M}\sum_{i=1}^{N} X_{i,j} \le 1, \quad \forall T_j \in T, \quad i \in [1,N], \quad j \in [1,M], \\ C_4: &\quad t_{i,j} \in [t_{s,j}, t_{e,j}], \quad \forall T_j \in T, \quad i \in [1,N], \quad j \in [1,M]. \end{aligned}$$

The objective function is subject to the following constraints: C_1 specifies the binary variable $X_{i,j}$ that denotes whether UAV U_i is assigned to task T_j; it is 1 if U_i is assigned to T_j, and 0 otherwise. C_2 denotes the maximum number of tasks L_i that can be covered by U_i. C_3 ensures that each task T_j can be performed by at most one UAV U_i. C_4 specifies the time window within which task T_j must be executed, where $t_{s,j}$ is the earliest start time, and $t_{e,j}$ is the latest end time.

These formulations are general and are widely used to represent optimization problems in multi-robot and multi-UAV systems. They are presented as a conceptual framework rather than being adapted from a specific source. However, similar formulations can be found in foundational studies, such as those by [25,26], which provide a basis for these optimization representations.

Example

To demonstrate the application of the proposed mathematical model, consider a scenario with three UAVs (U_1, U_2, U_3) and five tasks (T_1, T_2, T_3, T_4, T_5). The objective is to maximize the overall Benefit and minimize the total Cost of the task allocation while adhering to the model's constraints.

- Scenario Setup

  The Benefit of assigning UAV U_i to task T_j is given by the following:

  $$\text{Benefit}(T_j) = \text{Reward}(T_j),$$

  where $\text{Reward}(T_j)$ represents the value of completing task T_j.
  The Cost of assigning UAV U_i to task T_j is the distance between U_i and T_j:

  $$\text{Cost}(U_i, T_j) = \text{Distance}(U_i, T_j).$$

- Parameters

  Rewards for tasks (T_1, T_2, T_3, T_4, T_5): $[50, 40, 60, 30, 20]$.
  Distances from UAVs to tasks (T_j) (in arbitrary units):

	T_1	T_2	T_3	T_4	T_5
U_1	10	15	20	25	30
U_2	20	10	25	15	35
U_3	30	25	15	10	20

  Maximum tasks per UAV: $L = 2$.

- Optimization Calculation Using the model's objective functions:

  $$\text{Maximize:} \quad \sum_{j=1}^{5} X_{i,j} \cdot \text{Reward}(T_j),$$

  $$\text{Minimize:} \quad \sum_{i=1}^{3} \sum_{j=1}^{5} X_{i,j} \cdot \text{Distance}(U_i, T_j),$$

  subject to:

  Each UAV can be assigned up to two tasks: $\sum_j X_{i,j} \leq 2$.
  Each task can only be assigned to one UAV: $\sum_i X_{i,j} = 1$.

- Result:

  The resulting task allocation: U_1: (T_1, T_2), U_2: (T_4) and U_3: (T_3, T_5)
  The calculated total distances are as follows:

$$\begin{aligned} U_1: &\quad 10 + 15 = 25, \\ U_2: &\quad 15, \\ U_3: &\quad 15 + 20 = 35. \end{aligned}$$

  Total distance (cost): $25 + 15 + 35 = 75$.
  Total benefit: $50 + 40 + 30 + 60 + 20 = 200$.

Figure 1 illustrates the task-UAV assignment result, showing how tasks are allocated to UAVs based on the objective function.

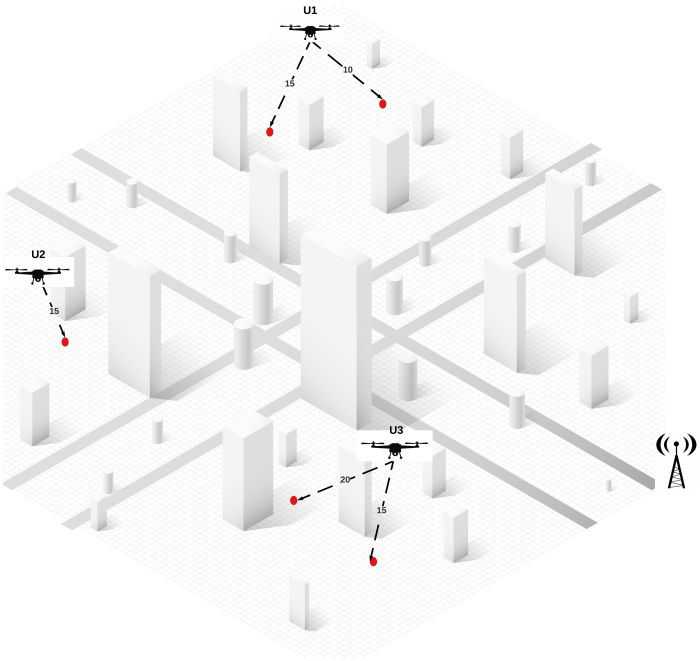


Figure 1. Illustration of the task assignment result for the example provided.

4. Multi-UAV Dynamic Task Assignment Solutions

The dynamic task assignment problem is an NP-hard combinatorial optimization problem [7,22,27]; therefore, finding an efficient solution to the problem is challenging. Several researchers have investigated the multi-UAV dynamic task assignment problem and have provided a wide range of solutions. This review covers literature studies of multi-UAV task assignment in dynamic environments from 2013 to 2024 and presents them according to the most common broad classification of dynamic task assignment approaches that include market-based, optimization-based, and cluster solutions.

To systematically compare these studies, comparison tables are presented that focus on solution characteristics, providing an overview of how different algorithms manage task allocation, including their control paradigms, UAV participation, reassignment strategies, dynamic event handling, and communication structures. The tables categorize the control paradigm as centralized, distributed, or hybrid, reflecting the coordination method for successful mission completion. UAV participation specifies whether all UAVs, none, or a subset are involved in task reassignment after dynamic events. Reassignment strategies are classified as none, partial, or full, based on how previously allocated tasks are adjusted to accommodate new events. Communication topology describes the structure and pro-

tocols used for UAV interaction, highlighting whether it is explicitly defined. Scalability categorizes dataset sizes as small (1–10 tasks), average (11–100 tasks), or large (100+ tasks). Robustness highlights the dynamic events the algorithms can handle, while responsiveness evaluates the consideration of time in the solution, either as an objective or performance metric. Time is classified as computation (decision-making time), execution (task completion time), waiting (delays before initiation), or converging (stabilization after changes). If time is not considered, the solution is marked as "Time Not Considered". These comparisons offer insights into the algorithms' ability to handle various dataset sizes, dynamic events, and time-based performance metrics.

Figure 2 shows the classification of multi-UAV dynamic task assignment solutions according to their classes.

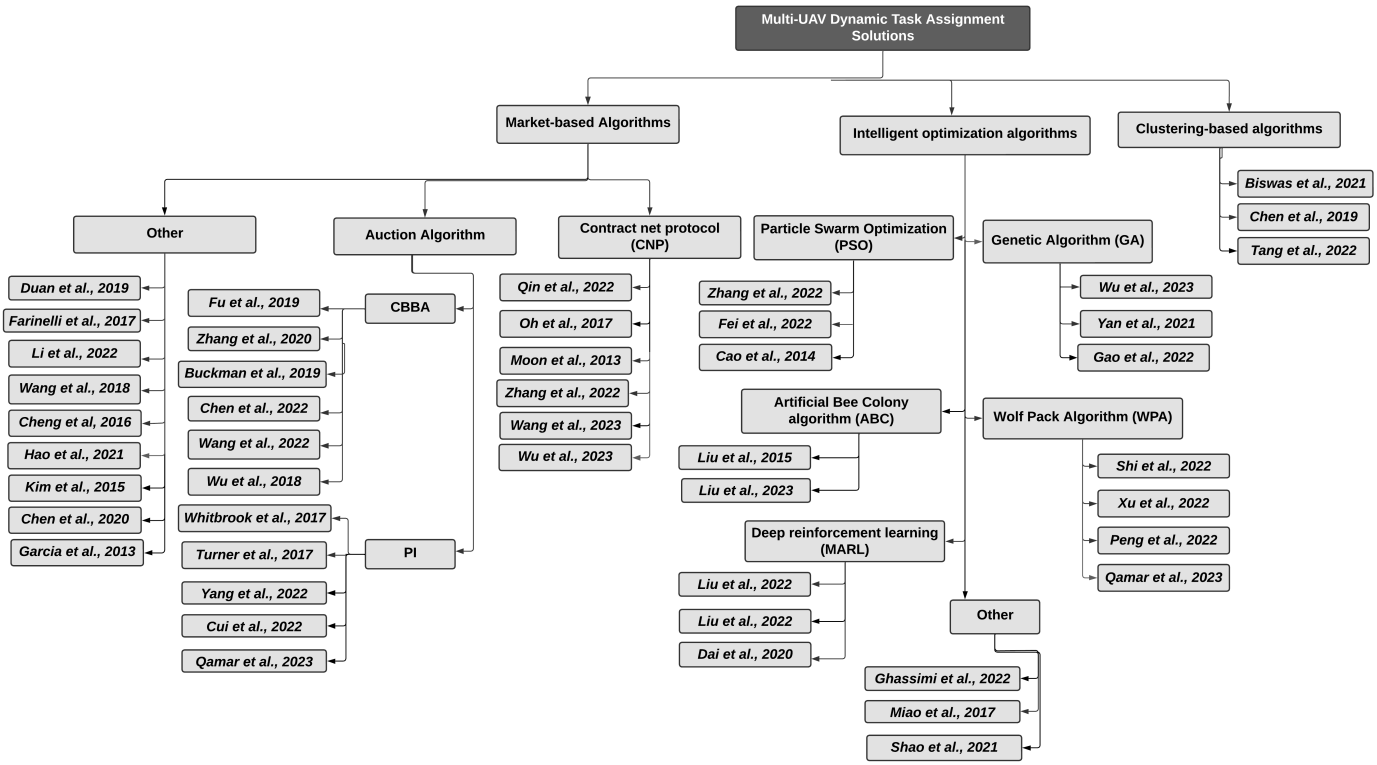


Figure 2. Multi-UAV dynamic task assignment solution classifications [6,7,18,19,27–68].

4.1. Market-Based Algorithms

Market-based algorithms are the most common approaches used for solving the multi-UAV task assignment problem due to their distributed behavior, which is suitable for the distributed structure of multi-UAV systems. This approach is inspired by the market trading–auction concept, which provides a communication mechanism between agents/UAVs to coordinate information exchange based on trading rules [69,70]. In multi-UAV systems, the auction process consists of a set of UAVs that bid for tasks based on their capabilities. The bidder UAVs negotiate with each other until they reach an agreement on the winning UAV that has the highest bidding, where the bidding is calculated based on optimizing a predefined objective function [71]. Common methods based on market mechanisms mainly include the auction algorithm and contract net protocol (CNP). The auction algorithms [72] assign UAVs to tasks by simulating the auction process in reality, which allows UAVs to bid for tasks and negotiate until one UAV with the highest bidding is assigned to the task. The following sections will go through different Market-based algorithms that handle multi-UAV task assignment problems in dynamic environments.

4.1.1. Consensus-Based Bundle Algorithm (CBBA)

The CBBA, proposed by Choi [73], is considered a multi-agent task allocation auction algorithm. The CBBA has two phases: the task inclusion phase, which uses a greedy algorithm to build a bundle of tasks in each UAV, and the consensus phase, which resolves the conflict between them. Different papers have extended the CBBA to address the multi-UAV dynamic task assignment problem. Fu et al. [28] proposed a task and resource dynamic assignment algorithm, extending the CBBA framework that uses a task sequence mechanism. The objective was to minimize UAV path lengths and maximize total task rewards by enabling UAVs with overlapping idle periods to bid and negotiate for conflict-free task assignments, inserting new tasks into an idle period without disrupting existing allocations. In simulations involving 14 UAVs with 4 original and 2 new tasks, the approach effectively minimized UAV path lengths, maximized total task rewards, and efficiently reassigned tasks and resources in a distributed manner. Compared to the baseline CBBA, the proposed solution also reduced computation time and communication overhead.

Zhang et al. [29] extended the CBBA framework with partial reassignment and asynchronous task allocation to address real-time challenges, minimizing both UAV flight distance and communication load. The algorithm released low-priority tasks for reallocation, ensuring efficient resource distribution. Simulations with 8 UAVs, 32 original tasks, and 4 new tasks demonstrated a reduction in task completion time, communication steps, and calculation time compared to the standard CBBA. The asynchronous mechanism allowed real-time performance with reduced network traffic, integrating new tasks smoothly without major delays.

Buckman et al. [30] proposed CBBA with partial replanning (CBBA-PR) to maximize time-discounted rewards by allowing a subset of UAVs to reallocate low-priority tasks alongside newly discovered ones to reduce communication rounds and runtime. The algorithm was tested with 8 UAVs performing 80 original and 8 new tasks. It demonstrated faster convergence than full reset strategies while maintaining higher solution quality than no-reset strategies. The CBBA-PR approach minimized disruption to ongoing tasks, balancing fast response times and task coordination in dynamic environments.

Chen et al. [27] introduced CBBA with local replanning (CBBA-LR), focusing on maximizing time-discount task rewards and minimizing the distance between UAVs and task locations. The algorithm allows UAVs with available resources to perform replanning for tasks with overlapping time windows and nearest distances to new tasks. This strategy minimized the number of involved UAVs while quickly integrating new tasks. Simulations with 25 UAVs, 100 original tasks, and 5 new tasks showed that CBBA-LR achieved better score increments and faster convergence compared to other reset strategies, offering high-quality task reassignment with lower computational complexity and quicker response times.

Wang et al. [31] proposed an improved CBBA with a two-tier task bidding mechanism to maximize the number of tasks performed and minimize task execution time. It adjusts task allocation sequences and re-planned tasks within specific time windows to integrate new tasks smoothly. Tested on 1 to 18 UAVs with 14 original and 2 new tasks, the approach enhanced task assignment efficiency and ensured timely task completion compared to CBBA and CNP. Although it improved performance, the solution incurred higher runtime costs.

Wu et al. [32] developed a distributed task allocation method using a consensus algorithm and an online cooperative strategy aimed to maximize total return value based on UAV trajectories and task execution time. The approach also employs a local chain communication mode, minimizing bandwidth requirements. In simulations involving 6 UAVs with 10 original and 7 new tasks, the algorithm improved total return value and task success

rates while reducing communication demands. Compared to Prim and CBBA, it achieved faster task completion and maintained higher success rates under uncertainty, with a 14% reduction in success rate compared to 28% in offline approaches. The solution efficiently handled new task appearances without requiring high-bandwidth communication.

Table 2 compares the characteristics of various CBBA extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. In studies extending the CBBA (consensus-based bundle algorithm) for multi-UAV dynamic task assignment, there are notable strengths and challenges across the implementations. The advantages of these extensions generally lie in improving task allocation efficiency and reducing communication overhead. For instance, Fu et al. [28] achieved greater efficiency by dynamically assigning new tasks without impacting pre-allocated ones, while Zhang et al. [29] introduced real-time partial reassignment and asynchronous task allocation, enhancing responsiveness. Buckman et al. [30] reduced the number of UAVs involved in reallocation, minimizing communication burden and execution time. Extensions such as CBBA-LR by Chen et al. [27] and Wang et al. [31] introduced mechanisms to handle dynamic task allocation events, focusing on quick responses and maximizing task rewards. However, the challenges include trade-offs in communication complexity and performance consistency. While some approaches reduce communication, they may lead to suboptimal task allocations (e.g., partial resets), while others that optimize rewards or performance tend to increase running time, as seen in the work by Wang et al. [31]. Overall, these extensions demonstrate a balance between achieving higher efficiency and responsiveness, yet they also expose the need to manage increased communication loads or execution time, depending on the strategy adopted.

Table 2. Comparison of solution characteristics for multi-UAV dynamic task assignment extending the consensus-based bundle algorithm (CBBA). (Paradigm: distributed).

Ref	Year	Participating UAVs	Type of Reassignment	Communication Topology	Scalability	Robustness Against Dynamic Events	Responsiveness
[28]	2019	All	No reassignment	No	Small	Appearance of new tasks + loss of existing tasks	Time-based (Computation)
[29]	2020	All	Partial	Yes	Small	Change of original tasks	Time-based (Execution)
[30]	2019	Partial	Partial	No	Average	Appearance of new tasks	Time-based (Completion + Calculation)
[27]	2022	Partial	Partial	No	Average	Appearance of new tasks	Time-based (Waiting)
[31]	2022	All	All	Yes	Average	Appearance of new tasks	Time-based (Waiting)
[32]	2018	All	No reassignment	Yes	Average	Appearance of new tasks	Time-based (Execution)

4.1.2. Performance Impact Algorithm (PI)

Zhao et al. [74] proposed a task allocation method known as the Performance Impact (PI) algorithm as an extension of the CBBA algorithm, which can allocate tasks in time-limited applications. PI is a distributed task allocation algorithm that comprises two phases. It considers the cost of assigning a task and also takes into account the impact of including a task in other tasks that are already allocated to each UAV. The PI has also been enhanced in different papers to handle the multi-UAV dynamic task assignment problem.

Turner et al. [33] proposed the PI-MaxAss algorithm, an extension of the performance impact (PI) algorithm, to maximize the number of allocated tasks in distributed multi-robot systems under strict time constraints. The algorithm shifts task assignments among vehicles to create feasible time slots for unassigned tasks without re-running the entire process. In simulations with 14 UAVs and 64 tasks, PI-MaxAss improved task allocation by up to 20% compared to CBBA and PI-MinAvg algorithms, although it required more iterations to converge.

Whitbrook et al. [19] extended the PI algorithm with two modules to address local minima and static structures by enabling dynamic rescheduling and improving search space exploration through a soft-max action–selection procedure. The solution aimed to minimize the mean-time cost of servicing tasks. Simulations with 16 UAVs, 96 original tasks, and 4 new tasks showed a 9% improvement in task completion time compared to CBBA. The rescheduling module enabled real-time adaptation but increased run times, especially for larger problem sizes.

Yang et al. [7] introduced a distributed reassignment framework that determines processing strategies based on the types of dynamic events and includes a partial reassignment algorithm (PRA). The PRA features an incremental sub-team formation mechanism allowing a subset of UAVs to respond to dynamic tasks and a partial releasing mechanism that deletes specific tasks to improve reallocation success rates. Additionally, they propose a modified inclusion phase to maximize assignment (MIP-MA) within the PRA to maximize the number of task allocations. The framework aimed to maximize total profit by considering both task scores and travel time costs. Simulations with 8 UAVs, 32 original tasks, and 2 new tasks showed that the PRA allocated 85% of new tasks while reducing communication and runtime compared to the PI and CBBA algorithms. The framework performed efficiently across various dynamic scenarios, such as task additions and removals.

Cui et al. [34] proposed a distributed allocation with time windows (DATW) algorithm to minimize the average task completion time while considering complex time constraints. The algorithm integrates time window constraints into a PI-based approach and uses a three-phase strategy—task inclusion, conflict resolution, and reallocation. By correlating task significance with time windows, tasks are completed early within their validity intervals without increasing communication burdens. In simulations with 16 UAVs and 48 tasks, DATW achieved up to an 18% increase in success rate compared to E-CBBA. The method also improved task allocation under varying task-to-UAV ratios, although larger task sets required more iterations and communications to reach consensus.

Qamar et al. [35] introduced the compromised dynamic PI (CDPI) algorithm to maximize task allocations and improve task prioritization in dynamic search-and-rescue scenarios. CDPI integrates the VIKOR method for multi-criteria decision-making, prioritizing tasks effectively. The algorithm was further extended to CDPI for dynamic scenarios, where new tasks are introduced during execution. Simulations with 6 UAVs, 12 original tasks, and 2 new tasks demonstrated a 22% increase in task allocations and a 28.57% improvement in task prioritization compared to the baseline PI algorithm. The algorithm also handled dynamic conditions efficiently, reallocating tasks with reduced costs.

Table 3 compares the characteristics of various PI extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. The PI algorithm and its extensions offer several advantages and challenges for multi-UAV dynamic task assignment. One of the key strengths is its ability to handle time-sensitive applications by considering the impact of new tasks on already assigned ones, leading to more efficient task allocation,

as demonstrated by Turner et al. [33], who reduced execution time and increased the number of allocated tasks. Whitbrook et al. [19] and Yang et al. [7] further enhanced real-time rescheduling and reassignment efficiency, reducing communication overhead and improving scalability in dynamic environments. However, some limitations include increased complexity and communication requirements in certain scenarios, such as the high number of iterations needed in PI-MaxAss, which can lead to longer convergence times. Additionally, the algorithm's focus on maximizing task allocation can sometimes result in trade-offs, such as slightly reduced solution quality or higher running times in complex dynamic scenarios, as seen in the work by Qamar et al. [35].

Table 3. Comparison of solution characteristics for multi-UAV dynamic task assignment extending the performance impact (PI) algorithm. (Paradigm: distributed; communication topology: no; scalability: average).

Ref	Year	Participating UAVs	Type of Reassignment	Robustness Against Dynamic Events	Responsiveness
[33]	2018	All	Partial	Appearance of new tasks + change original tasks + adding new UAVs + loss of UAVs	Time-based (Waiting)
[19]	2017	All	Partial	Appearance of new tasks	Time-based (Execution)
[7]	2022	Partial	Partial	Appearance of new tasks + change original tasks	Time-based (Waiting)
[34]	2022	All	No reassignment	Change original tasks	Time-based (Completion)
[35]	2023	All	No reassignment	Appearance of new tasks	Time-based (Execution)

4.1.3. Contract Net Protocol Algorithm (CNP)

The contract net protocol (CNP), introduced by Reid G. Smith in 1980 [75], is a task-sharing protocol commonly used in multi-agent systems for task allocation among autonomous agents. It closely resembles sealed auction protocols. In CNP, a manager agent proposes a task to multiple agents, who then submit bids or proposals. The manager selects one of the proposals to assign the task. Additionally, the assigned task can be further divided and subcontracted to other agents if necessary. Different papers have enhanced the CNP algorithm. Qin et al. [6] proposed a distributed grouping cooperative dynamic task assignment method based on an extended contract network protocol (CNP) for UAV swarms performing reconnaissance-and-attack tasks on multiple targets to maximize UAV rewards and minimize penalties based on fuel consumption and air range. This approach enhanced real-time task performance in complex combat scenarios with sudden target appearances. Simulations with 11 UAVs, 12 original tasks, and 4 new tasks showed that the strategy outperformed ACO, requiring fewer computational resources and less time, validating its effectiveness for dynamic environments.

Oh et al. [36] proposed market-based decentralized algorithms extending the CNP to maximize mission scores by prioritizing tasks and considering completion times. The algorithm includes two decentralized algorithms where a coalition leader is elected to eliminate continuous synchronization among agents. Simulations with 10 UAVs, 30 original tasks, and 2–3 new tasks showed better task completion compared to GDAP, though more communication was required due to frequent resets. Modified GDAP reduced communication costs by avoiding consensus processes, and enhancing scalability and effectiveness in dynamic environments.

Moon et al. [37] proposed a negotiation-based task assignment algorithm using the CNP, aiming to minimize UAV travel distances. In this approach, each UAV selects tasks based on the lowest cost, with conflicts resolved by awarding tasks to the UAV with the lowest cost. Tested with 3 UAVs and 10 tasks, the algorithm demonstrated faster computation than MILP, ensuring successful mission completion and making it suitable for dynamic environments with low computational costs.

Zhang et al. [38] developed a hybrid CNP consisting of buy–sell, swap, and replacement contracts to optimize total profit, minimize time consumption, and balance task scheduling. Simulations with 3 UAVs, 10 tasks, and one new task showed that the hybrid CNP reduced rescheduling time to 0.003 s and increased rewards from 202 to 272 by reallocating low-reward tasks in comparison to basic CNP-based methods. The approach proved particularly effective in dynamic scenarios with emergencies and platform failures.

Wang et al. [39] introduced the TS-DTA algorithm, a two-stage distributed task assignment method designed to minimize mission completion and flight times while managing new targets and UAV failures in reconnaissance-and-attack environments. Simulations with 8 UAVs, 20 original tasks, and 5 new tasks showed that TS-DTA reduced communication burdens while achieving a high success rate for task reassignments. It outperformed PSO and CNP algorithms, improving task allocation efficiency by 20% in scenarios involving new targets or UAV damage.

Wu et al. [40] proposed a hierarchical mission replanning strategy as an adaptation of the contract net protocol for multiple UAV formations, focusing on minimizing path length and resource usage. The method used a rapid-exploration random tree algorithm for path planning and task reallocation, ensuring tasks were assigned to the most capable UAV formations and reducing the necessity for extensive online computation, allowing UAV formations to swiftly adjust to changes and efficiently fulfill their objectives. Simulations with 10 UAVs, 18 original tasks, and 2 new tasks showed that the hierarchical strategy reduced mission completion time by balancing task loads and UAV capabilities, enhancing performance in dynamic environments.

Table 4 compares the characteristics of various CNP extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. The contract net protocol (CNP) algorithm and its extensions have notable strengths and weaknesses for multi-UAV dynamic task assignment. One of the key advantages is the algorithm's flexibility in handling dynamic environments, allowing efficient task allocation even in complex scenarios, as seen in the work by Qin et al. [6] and Moon et al. [37]. The hierarchical and coalition-based approaches reduce the computational burden and allow for real-time task rescheduling, particularly effective in dynamic and unpredictable environments. Furthermore, the use of hybrid contracts, such as those proposed by Zhang et al. [38], enables optimization of task allocation with minimal communication overhead. However, one drawback is that some extensions, like Oh et al. [36], face challenges in scenarios with frequent network disconnections, where continuous communication between agents becomes difficult. Additionally, in cases where there is no reassignment, task efficiency can be limited when new tasks appear or unexpected events occur, as seen in the work by Wu et al. [40]. Despite these limitations, the CNP and its variations provide scalable solutions with significant performance improvements in dynamic task environments.

Table 4. Comparison of solution characteristics for multi-UAV dynamic task assignment extending the contract net protocol (CNP) algorithm. (Participating UAVs: all).

Ref	Year	Paradigm	Type of Reassignment	Communication Topology	Scalability	Robustness Against Dynamic Events	Responsiveness
[6]	2022	Centralized-Distributed	Partial	Yes	Average	Appearance of new tasks + loss of UAVs	Time-based (Execution)
[36]	2017	Distributed	No reassignment	Yes	Average	Network disconnections	Time-based (Completion)
[37]	2013	Distributed	No reassignment	No	Small	Appearance of new tasks	Time-based (Completion)
[38]	2022	Distributed	Partial	No	Small	Appearance of new tasks + loss of UAVs	Time-based (Execution)
[39]	2023	Distributed	Partial	Yes	Average	Appearance of new tasks + change original tasks + loss of UAVs	Time-based (Completion)
[40]	2023	Distributed	No reassignment	No	Average	Appearance of new tasks + change original tasks	Time-based (Completion)

4.1.4. Other Extensions of Market-Based Algorithms

Other literature papers have extended other market-based algorithms. Based on the distributed auction algorithm, Duan et al. [18] proposed a hybrid two-stage auction algorithm based on a hierarchical decision mechanism and centralized-distributed auction structure. The algorithm aims to maximize mission payoff, minimize cost (fuel, time, and distance), and optimize task coverage. In the first stage, the algorithm selects a task that urgently needs to be performed and then generates a reasonable auction sequence based on the hierarchical decision mechanism instead of a random sequence, as in previous works. Then, related UAVs bid for this task according to the objective function, which considers the association cost between tasks and the penalty factor for balancing the use of UAVs. The procedure is repeated until all tasks are assigned. Simulations with 8 UAVs, 20 original tasks, and 4 new tasks showed that the hybrid auction method outperformed CAA, CBBA, and DTAP algorithms by improving total mission payoff and task completion speed, due to the adaptive penalty and task coverage factors.

Farinelli et al. [41] addressed multi-robot patrolling and introduced two approaches for dynamic task assignment: DTA-Greedy and DTAP (a market-based strategy using sequential single-item auctions). The objectives were to minimize global idleness and travel costs. The first technique, DTA-Greedy, performs a greedy maximization decision on the objective function by considering the current idleness and receiving one from other robots. The other technique is inspired by sequential single-item auctions, where robots allocate one task at a time while considering the other allocated tasks. Simulations with 8 UAVs and 163 tasks demonstrated that DTAP outperformed RAND and SEBS in reducing idleness and achieving smoother task execution with fewer conflicts. DTA-Greedy was less effective but still improved on simpler methods like RAND, CR, and HCR.

Li et al. [42] proposed an online shared memory auction algorithm to maximize total payoffs, which resets some of the allocated tasks and re-auctions them together with the new tasks without completely resetting the existing assignment. Simulations with 10 UAVs, 60 original tasks, and 5 new tasks showed that the online auction method achieved higher payoffs than sequential auctions and reduced auction rounds compared to offline auctions, resulting in faster task assignment.

Wang et al. [43] developed an auction-based dynamic task allocation algorithm for heterogeneous manned-unmanned aerial vehicle teamwork (MUAVT), focusing on maximizing task success probability while considering agent capacities and task distances. The unassigned tasks are initially updated and reordered according to their priority. Using the

extended auction algorithm, the agents then bid on the unallocated task while considering as constraints of time and resources. Then, in order to resolve conflicts between agents, the priority-based task coordination module is employed. Once the success probability exceeds the threshold or there are no agents available, the bidding process ends. In scenarios with 1 MAV, 4 UAVs, and 3 original tasks, the algorithm efficiently reassigned tasks in response to changing priorities and conditions, ensuring mission success despite environmental threats.

Cheng et al. [44] presented an auction-based multi-constraint task allocation algorithm to maximize UAV rewards and minimize completion costs (including fuel, sensor, and danger factors). The key idea is to arrange these constraints in a fixed sequence, and then update the cost from the first layer to the last until all constraints have been satisfied. The task reward is considered after all constraints have been solved. When a task is identified, only those UAVs that can meet the requirements under all four constraints are selected to accomplish the task. Simulations with 30 UAVs, 50 tasks, and 5 danger areas showed that average fuel costs decreased as more UAVs completed tasks. Results also demonstrated that task success rates stabilized with an increased number of UAVs, even as task volumes grew.

Hao et al. [45] proposed a distributed–centralized dynamic task allocation framework combining a distributed election algorithm and a centralized fast online target assignment algorithm to minimize the total tracking costs of multiple moving targets. As a first step, a distributed leader election algorithm is designed in order to determine a leader, which acts as the central node for the acquisition of information about UAVs and targets, and for the allocation of tasks. Secondly, a fast online target assignment algorithm is proposed to dispatch UAVs to track moving targets. Simulations with 1200 UAVs and 800 tasks showed that the framework computed solutions within 10 s for fewer than 1000 UAVs. It effectively balanced energy consumption and target priority to minimize tracking costs.

Kim et al. [46] introduced a resource welfare-based task allocation strategy using an auction-like protocol to maximize total rewards in search and attack missions. This method allows a UAV team to efficiently manage resources by balancing depletion rates, thereby ensuring a greater availability of UAVs to adaptively respond to dynamic situations. Simulations with 5 UAVs and 30 tasks showed that the welfare-based method outperformed instantaneous task allocation, achieving 10% to 29.3% higher total rewards across different scenarios.

García et al. [47] proposed a simple auction-based strategy designed for scalability into distributed multi-sub-auctions to handle the increasing complexity of dynamic task assignments for large multi-robot teams. The core aim is to minimize travel distances and demonstrate that this simple auction framework enables linear computational cost growth relative to the team size, facilitating the resolution of intricate assignment challenges under dynamic conditions through a hierarchical sub-auction approach. Simulations with 24 UAVs and 6 tasks showed that auction times increased linearly with team size, demonstrating scalability even in complex, hierarchical setups.

Cheng et al. [48] developed NECTAR, an agent-based task allocation algorithm using an enhanced auction mechanism to minimize execution time and maximize task success rates. NECTAR facilitates rapid response to dynamic tasks with minimal communication overhead, enabled by collaborative agents acting based on local statuses, ensuring continuous task assignment. Simulations with 150 UAVs showed that NECTAR achieved over 80% task success in most scenarios, outperforming the RANDOM baseline. NECTAR's higher execution time ratio indicated more efficient task execution during UAV flights.

Table 5 compares the characteristics of the various other extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against

dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. Other market-based algorithms for multi-UAV dynamic task assignment offer several advantages and challenges. On the positive side, some of these algorithms efficiently utilize local information exchange and UAV preferences to produce robust solutions that can adapt to uncertain, dynamic environments. These approaches are highly scalable, handling increasing task complexity with mechanisms like re-auctioning and adaptive bidding strategies, as seen in the work by Li et al. [42] and Wang et al. [43], which ensure good-quality task assignments even as the input size grows. Additionally, algorithms like NECTAR [48] achieve high task success rates with minimal communication overhead, making them effective for real-time task reallocation. However, the reliance on local communication status can lead to delays in obtaining solutions, particularly in cases with frequent network disconnections or high communication loads, as seen in the work by Farinelli et al. [41]. Moreover, while market-based approaches generally reduce computational complexity, excessive computation can occur in scenarios with large task sets or complex reallocation strategies, potentially slowing down the task assignment process. Overall, these algorithms strike a balance between efficiency and adaptability but are sensitive to communication and computation bottlenecks. In general, market-based algorithms have many advantages, including the ability to achieve an efficient solution by utilizing local information exchange and agent/UAV preference, the robustness against uncertain changes in dynamic environments, and the scalability of the algorithms to obtain a good-quality solution when the size of the input increases. However, market-based approaches depend heavily on local communication status, which may lead to delays in obtaining a solution and can involve excessive computation.

Table 5. Comparison of solution characteristics for multi-UAV dynamic task assignment extending other market-based algorithms.

Ref	Year	Paradigm	Participating UAVs	Type of Reassignment	Communication Topology	Scalability	Robustness Against Dynamic Events	Responsiveness
[18]	2019	Centralized-Distributed	Partial	No reassignment	No	Average	Appearance of new tasks + loss of UAVs	Time-based (Waiting)
[41]	2017	Distributed	All	No reassignment	No	Large	Change original tasks	Time-based (Waiting)
[42]	2022	Distributed	All	Partial	No	Average	Appearance of new tasks	Time-based (Converging)
[43]	2018	Centralized-Distributed	All	Partial	No	Small	Appearance of new tasks + change original tasks + loss of UAVs	Time Not Considered
[44]	2016	Distributed	All	No reassignment	No	Average	Appearance of new tasks	Time-based (Completion)
[45]	2021	Centralized-Distributed	All	No reassignment	No	Large	Appearance of new tasks + loss of UAVs	Time-based (Computation)
[46]	2015	Distributed	Partial	No reassignment	No	Average	Appearance of new tasks + loss of UAVs	Time Not Considered
[47]	2013	Centralized-Distributed	Partial	No reassignment	Yes	Small	Change original tasks	Time-based (Computation)
[48]	2020	Centralized-Distributed	All	No reassignment	Yes	Large	Appearance of new tasks	Time-based (Execution)

4.2. Intelligent Optimization Algorithms

Intelligent optimization algorithms are random search methods with a mathematical formulation inspired by natural phenomena or social behavior [22]. These algorithms work to find an optimal solution out of a set of feasible solutions that are restricted by some constraints. The optimal solution is selected based on a predefined objective function that measures solution quality [3]. Optimization-based algorithms can be classified according to their behavior into evolutionary algorithms and swarm intelligence algorithms. Evolutionary algorithms are inspired by the biological evolution process, and one of the most commonly used evolutionary algorithms in task assignment problems is the genetic

algorithm. The swarm intelligence algorithms are inspired by the biological behavior of groups in nature, where a set of individuals (candidate solutions) combine to form a group and optimize their solution quality by exchanging information with each other. Genetic algorithm (GA) [76], ant colony optimization (ACO) [77], and particle swarm optimization (PSO) [78] are the most commonly used swarm intelligence algorithms.

4.2.1. Genetic Algorithm (GA)

A genetic algorithm (GA) is a metaheuristic inspired by the process of natural selection and is part of the broader category of evolutionary algorithms (EA). GAs are frequently applied to produce high-quality solutions for optimization and search problems by utilizing biologically inspired mechanisms such as mutation, crossover, and selection [76]. By reviewing the literature, we found that different optimization-based approaches have been extending genetic algorithms to solve multi-UAV dynamic task allocation problems. Wu et al. [49] proposed a hybrid task assignment method using a distributed genetic algorithm (DGA) combined with an extended CBBA for time-critical tasks in dynamic disaster relief scenarios. The method aims to maximize task-assignment rewards and minimize operational time and UAV paths. The DGA optimizes operation time and enhances communication among UAVs, enabling efficient reallocation when new tasks arise. When a new task emerged, some UAVs were selected to reassign their current allocations, and only the last task was set to be released to handle the new task. Simulations with 5 UAVs and 15 tasks showed that the algorithm outperformed PI, HRCA, and CGBA in quality index score, with superior rewards and operation times compared to HRCA, achieving reallocation in 0.1 s.

Yan et al. [50] developed a heterogeneous UAV cooperative task allocation algorithm based on genetic and improved CNP algorithms to maximize task rewards and minimize UAV path lengths. The algorithm improves task balance, distribution efficiency, and dynamic planning in response to UAV damage or new targets. As a result of unexpected circumstances, such as damage to a UAV or a new target, first, calculate the load rate of each UAV, and the UAV with the highest load rate will be responsible for bidding on the target. To maximize the overall effectiveness of the UAV, the proposed algorithm rearranges the execution order of the UAV when it acquires a new task. Simulations with 6 UAVs, 20 tasks, and 2 damaged UAVs showed the algorithm exceeded traditional CNP performance by balancing task allocation, reducing execution times, and raising effectiveness scores from 843.41 to 1326.18, demonstrating resilience in dynamic conditions.

Gao et al. [51] introduced a multi-objective framework combining NSGA-II for initial task optimization and a real-time re-assignment mechanism via CNP to maximize destroyed task values and minimize the costs of damaged UAVs. Designed for dynamic environments with UAV damage or emerging targets, the CNP's bidding system enables agile redistribution of tasks. Simulations with 15 UAVs and 100 tasks showed that the proposed method solved task reassignment faster than MOPSO, with higher solution scores and shorter CPU runtime, balancing rapid response with slight global optimization sacrifices.

Table 6 compares the characteristics of the various other extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. Genetic algorithms (GAs) for multi-UAV dynamic task assignment offer significant advantages and some limitations. The strengths include their adaptability and effectiveness in dynamic environments, as they can optimize task assignments in real-time, even when new tasks or unexpected events such as UAV damage occur. For example, Gao et al. [51] demonstrated

that their GA-based approach efficiently reassigns tasks faster than other benchmarks like MOPSO, allowing for agile and responsive operations. The ability of GAs to balance multiple objectives, such as minimizing operational time and maximizing rewards, as seen in the work by Wu et al. [49], showcases their versatility in optimizing task allocation. However, the main drawback is that these algorithms sometimes sacrifice global optimization for quicker, real-time solutions, which may not always yield the best overall outcome, particularly in complex scenarios. Additionally, the computational overhead associated with evolving solutions, as shown in the work by Yan et al. [50], can be significant, and balancing this with response time remains a challenge. Overall, while GAs are powerful for handling dynamic task allocation, they must be carefully tuned to manage trade-offs between solution quality and computational efficiency.

Table 6. Comparison of solution characteristics for multi-UAV dynamic task assignment extending the genetic algorithm (GA). (Paradigm: distributed; scalability: average).

Ref	Year	Participating UAVs	Type of Reassignment	Communication Topology	Robustness Against Dynamic Events	Responsiveness
[49]	2023	Partial	Partial	Yes	Appearance of new tasks	Time-based (Execution)
[50]	2021	All	No reassignment	No	Appearance of new tasks + loss of UAVs	Time-based (Execution)
[51]	2022	All	No reassignment	No	Appearance of new tasks + loss of UAVs	Time-based (Completion)

4.2.2. Particle Swarm Optimization Algorithm (PSO)

Particle swarm optimization (PSO) is a computational technique that seeks to optimize a problem by iteratively refining candidate solutions based on a specific measure of quality [78]. Different optimization-based approaches have been extending PSO to solve multi-UAV dynamic task allocation problems. Zhang et al. [38] proposed a hybrid A-QCDPSO algorithm integrating an improved discrete PSO with a market auction mechanism to maximize overall benefits and minimize threat and range costs. Particle subpopulations were dynamically divided according to their quality, altering the algorithm's topology. The market auction mechanism was applied during particle initialization and task coordination to generate high-quality particles. In simulations with 6 UAVs, 20 tasks, 2 emergencies with sudden UAV failure, and 3 new tasks, the A-QCDPSO algorithm outperformed QCDPSO and traditional contract network methods in task execution time and fitness, demonstrating enhanced efficiency in dynamic task allocations.

Fei et al. [52] introduced a dynamic task allocation framework using divide-and-conquer principles with spectral clustering (SC) and PSO to minimize delay penalties and energy consumption for UAVs in urban cargo distribution. The task allocation problem is decomposed into several sub-problems based on the divide-and-conquer principle, and an initial task allocation scheme is generated using the spectral clustering (SC) algorithm. Then, the tasks within each cluster are ranked using the PSO algorithm to obtain the optimal task allocation scheme. The task allocation mechanism is dynamically adjusted in the case of interruption of tasks in order to maintain high task completion rates and reduce UAV execution costs. Simulations with 6 UAVs and 100 tasks demonstrated effective task order optimization, lowering delays and energy usage.

Cao et al. [53] addressed dynamic battlefield task allocation with a mixed allocation method that divides the problem into group-level and member-level allocations, solved using an enhanced PSO-FSA and a distributed auction algorithm, streamlining time and communication complexity. The objective is to minimize total flight distance while maintaining low communication complexity. Simulations with 9 UAVs, 20 tasks, and 3 new tasks showed that this method reduced task allocation time and communication complexity

compared to distributed auction algorithms, though it had slightly lower stability due to negotiation conflicts. The approach balances robustness and autonomy, making it suitable for complex environments.

Table 7 compares the characteristics of the various other extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. The particle swarm optimization (PSO) algorithm, extended for multi-UAV dynamic task allocation, offers both strengths and weaknesses across different implementations. One of the key advantages is its adaptability to dynamic environments and ability to handle real-time requirements effectively. For example, Zhang et al. [38] introduced the A-QCDPSO, which excels in optimizing task allocation time and improving task execution efficiency, outperforming traditional algorithms in sudden UAV failure or emergency task scenarios. Similarly, Fei et al. [52] demonstrated PSO's strength in reducing energy consumption and maintaining high task completion rates in urban cargo distribution tasks, showcasing its effectiveness in large, complex problems. Cao et al. [53] extended PSO by integrating it with a distributed auction algorithm, significantly reducing communication complexity and improving time performance in battlefield scenarios. However, there are drawbacks. The stability of some methods, such as Cao's PSO-FSA approach, can be compromised by prolonged negotiation phases, leading to lower consistency in performance. Additionally, the reliance on local optimization in some cases, as with Fei et al. [52], means that the algorithms might not always guarantee globally optimal solutions, especially when comparisons with other benchmarks are lacking. Overall, PSO offers efficiency and scalability but can be hindered by stability issues and the complexity of certain dynamic scenarios.

Table 7. Comparison of solution characteristics for multi-UAV dynamic task assignment extending the particle swarm optimization algorithm (PSO). (Type of reassignment: no reassignment; scalability: average).

Ref	Year	Paradigm	Participating UAVs	Communication Topology	Robustness Against Dynamic Events	Responsiveness
[38]	2022	Distributed	All	No	Appearance of new tasks + loss of UAVs	Time-based (Converging)
[52]	2022	Distributed	None	No	-	Time-based (Waiting)
[53]	2014	Centralized-Distributed	Partial	Yes	Appearance of new tasks	Time-based (Completion)

4.2.3. Artificial Bee Colony Algorithm (ABC)

The artificial bee colony (ABC) algorithm is an optimization method that mimics the foraging behavior of honey bees and has been effectively applied to numerous real-world problems. ABC, introduced by Karaboga in 2005, is part of the swarm intelligence family of algorithms [79]. Numerous previous studies have extended ACO to address the problem of dynamic task assignment in multi-UAV systems. Liu et al. [54] introduced a dynamic task assignment approach using the artificial bee colony (ABC) algorithm to minimize total assignment costs, including execution and communication costs, in multi-agent cooperative systems. Each bee represents a task-agent combination, and task assignments are generated through a heuristic-based initialization. Simulations with 10 agents and up to 100 tasks showed that ABC outperformed GA and PSO in assignment cost, computation time, and scalability, particularly in complex, high-density task environments, demonstrating its effectiveness with lower computational effort.

Liu et al. [55] proposed Ex-MADDPG, an enhancement of the MADDPG algorithm tailored for maximizing task completion rates and minimizing task loss and communication overhead in UAV swarm target-attack missions. Inspired by bee colony dynamics, Ex-MADDPG supports large-scale swarms where UAVs independently identify and share target locations within limited communication ranges from adjacent units, and make collective decisions for varied task execution. In simulations with 56 UAVs and 28 tasks, Ex-MADDPG outperformed MADDPG and ms-MADDPG in task completion rate, decision time, and communication load. Real-world tests confirmed high performance in dynamic scenarios, assigning UAVs to tasks efficiently in real-time.

Table 8 compares the characteristics of the various other extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. The artificial bee colony (ABC) algorithm has shown several advantages and drawbacks when applied to multi-UAV dynamic task assignment problems. One of the key strengths of the ABC algorithm is its scalability and efficiency in handling complex, high-density task scenarios, as demonstrated by Liu et al. [54], where the algorithm outperformed both GA and PSO in terms of assignment cost and computation time. Its ability to minimize total costs and adapt to dynamic environments makes it well-suited for large-scale cooperative systems. Additionally, the ABC algorithm is robust in dynamic scenarios, effectively managing new task appearances, agent failures, and mission parameter changes. On the other hand, one limitation is that, like other swarm intelligence algorithms, it heavily relies on communication between agents, which may increase overhead in highly dynamic or information-limited environments, as seen in the work by Liu et al. [55]. The Ex-MADDPG variant inspired by bee colony behavior improves scalability and applicability in larger swarms, but the method's complexity might lead to increased communication loads when dealing with more intricate real-time scenarios. Despite these challenges, the ABC algorithm and its extensions have demonstrated strong performance in dynamic task environments, especially in minimizing computation effort and achieving higher task completion rates.

Table 8. Comparison of solution characteristics for multi-UAV dynamic task assignment extending the artificial bee colony algorithm (ABC). (Paradigm: distributed, participating UAVs: partial; communication topology: yes; scalability: average)

Ref	Year	Type of Reassignment	Robustness Against Dynamic Events	Responsiveness
[54]	2015	Full	Appearance of new tasks + loss of agent + changes in mission parameters	Time-based (Computation)
[55]	2023	No reassignment	Appearance of new tasks	Time Not Considered

4.2.4. Gray Wolf Algorithm

Gray wolf optimization (GWO) is a metaheuristic algorithm inspired by the leadership structure and hunting behavior of gray wolves, as introduced by Mirjalili et al. in 2014 [80]. Many studies have adapted GWO to tackle the challenge of dynamic task assignment in multi-UAV systems. Shi et al. [56] proposed the TABA method, a multi-UAV task assignment approach based on bionic algorithms (ant colony, bat, and gray wolf algorithms) to minimize calculation time, travel time, and distance traveled by UAVs. The method adapts task allocation based on the number of UAVs and tasks, if the number of UAVs is less than the number of tasks, the farthest UAVs will stop searching. If the number of UAVs is greater than the tasks, then the tasks will be only assigned to the nearby UAVs, which

reduces the computational time and the UAV traveled distance. Simulations with five tasks and three UAVs showed that TABA reduced UAV travel time by about one-third compared to baseline bionic algorithms, demonstrating lower complexity and high efficiency.

Xu et al. [57] introduced the multi-discrete wolf pack algorithm (MDWPA) to minimize resource consumption and maximize task execution efficiency in dynamic and complex environments. The MDWPA adapts wolf pack behaviors to improve accuracy and convergence in task assignments by discretizing the behaviors of wandering, calling, sieging, and incorporating new individual supplementation. Tested with 15 UAVs and 75 tasks, MDWPA outperformed PSO and GA in resource consumption and task completion time, achieving the shortest UAV flight distances and maintaining high solution accuracy and convergence speed.

Peng et al. [58] developed the binary wolf pack algorithm (BWPA) for UAV task allocation in uncertain environments to maximize UAV flight range and task efficiency. By using this method, the attacker establishes a dynamic allocation model while searching for and attacking higher-priority targets first, using a cost-effectiveness ratio and task execution time window. In simulations with 15 UAVs and 20 tasks, BWPA outperformed BPSO, BASFA, and GA, achieving greater task efficiency and lower computational costs, showcasing stability and adaptability in dynamic scenarios.

Qamar et al. [59] proposed the four-dimensional information gray wolf optimization (4DI-GWO) algorithm to minimize threat and flight costs while maximizing target destruction gains. The 4DI-GWO algorithm enhances task assignment by accounting for uncertainties in target locations, mission duration, and UAV survival probabilities. Simulations with 10 UAVs and 16 targets demonstrated faster convergence, superior global search, and higher task completion efficiency compared to GWO, PSO, and GA, highlighting its robustness in dynamic environments.

Table 9 compares the characteristics of the various other extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. The gray wolf optimization (GWO) algorithm and its various adaptations for multi-UAV dynamic task assignment demonstrate both advantages and challenges. One of the key strengths of GWO is its ability to efficiently handle complex and dynamic environments, as seen in the work by Shi et al. [56] and Xu et al. [57], where the proposed methods reduced UAV travel time and computational complexity while ensuring optimal task assignment. GWO-based approaches excel at minimizing resource consumption, achieving high task execution efficiency, and maintaining robust convergence speeds, as demonstrated in Peng et al. [58] and Qamar et al. [59]. These algorithms also outperform traditional methods like PSO, GA, and ACO in terms of convergence, flight distance, and task allocation efficiency, particularly in uncertain environments or battlefield scenarios where rapid adjustments are required. However, a notable drawback is that these approaches often depend on the specific dynamics of the problem, and some variants may not always guarantee the best performance under extreme uncertainty, as they might struggle with balancing global and local search capabilities. Additionally, while algorithms like 4DI-GWO [59] offer improved convergence speeds, the complexity of managing four dimensions of information could result in higher computational costs under certain circumstances. Overall, GWO extensions are highly effective for dynamic, real-time multi-UAV task allocation, but trade-offs between complexity and computational cost must be considered.

Table 9. Comparison of solution characteristics for multi-UAV dynamic task assignment extending the Wolf Algorithm. (Paradigm: distributed; type of reassignment: no reassignment; communication topology: no).

Ref	Year	Participating UAVs	Scalability	Robustness Against Dynamic Events	Responsiveness
[56]	2022	None	Small	-	Time-based (Computation + Waiting)
[57]	2022	All	Average	-	Time-based (Converging)
[58]	2022	All	Average	-	Time-based (Computation)
[59]	2024	None	Average	Change number of UAVs	Time-based (Converging)

4.2.5. Deep Learning Algorithm

Deep learning enables computational models with multiple processing layers to learn data representations at various levels of abstraction. These techniques have significantly advanced the state-of-the-art in dynamic task assignment for multi-UAV systems [81]. Liu et al. [60] proposed a coordinated dynamic task allocation (CDTA) strategy using multi-agent deep reinforcement learning (MARL) with a Q-network to minimize UAV-to-target distances in uncertain environments. MARL proposes requests and determines the responses of UAVs through the use of the coordinated network, and a Q-network is developed to estimate the expected return and to determine whether a responder will participate in the new dynamic task, reducing online computation burden and enhancing operation speed. Simulations with 10 to 180 UAVs and 10 dynamic tasks showed faster convergence and better task allocation efficiency than game theory-based and FTA approaches, demonstrating strong scalability as the number of UAVs increased.

Liu et al. [61] introduced SMADDPG, a multi-agent deep deterministic policy gradient algorithm for UAV task planning in a 3D dynamic environment to minimize travel distances and execution times. SMADDPG adeptly adjusts to varying operation scales by distinguishing between local and global network inputs, effectively managing the complexities of changing UAV cluster sizes. Simulations with agent scales of 6 and 12 UAVs showed that SMADDPG outperformed DDPG and MADDPG in task completion efficiency, convergence speed, and success rate, effectively managing dynamic conditions and maintaining high task success rates in both static and dynamic environments.

Dai et al. [62] explored multi-UAV task allocation for exploration and destruction missions using three approaches: auction-based, vacancy chain, and deep Q-learning (DQN), aiming to minimize travel costs and improve task distribution. Differences between the proposed algorithms are mainly related to the criteria considered when unallocated tasks are assigned to robots. In simulations with 10 UAVs and 40 tasks, the auction-based approach excelled in minimizing travel costs but led to load imbalance. The vacancy chain approach balanced the load and reduced completion times through dynamic task reallocation, though it incurred higher computational costs. DQN adapted well to dynamic environments, providing balanced results with slightly lower efficiency compared to the other methods.

Table 10 compares the characteristics of the various other extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. Deep learning algorithms have shown several advantages and challenges when applied to multi-UAV dynamic task assignment. One of the major strengths is their adaptability and scalability in dynamic and uncertain environments, as demonstrated by Liu et al. [60] and Liu et al. [61], where deep learning-based approaches like MARL and SMADDPG efficiently handled

dynamic task assignment with improved task completion rates, faster convergence, and strong scalability even as the number of UAVs increased. These algorithms can effectively manage both local and global complexities in dynamic environments, maintaining high task success rates and efficiently reallocating tasks when new ones appear or when UAVs fail. However, a drawback of these deep learning methods is that—compared to traditional algorithms like auction-based methods or game theory approaches—they can require higher computational resources, as seen in the work by Dai et al. [62]. Furthermore, while deep Q-learning can achieve task allocation without pre-designed rules, it may perform slightly less efficiently in terms of load balancing and overall task completion time compared to other heuristic methods like vacancy chains or auction-based approaches. Overall, deep learning models provide a flexible and scalable solution for dynamic task allocation, but they may need to balance computational overhead and efficiency, particularly in large-scale or complex environments.

Table 10. Comparison of solution characteristics for multi-UAV dynamic task assignment extending the deep learning algorithms (DLs). (Paradigm: distributed).

Ref	Year	Participating UAVs	Type of Reassignment	Communication Topology	Scalability	Robustness Against Dynamic Events	Responsiveness
[60]	2022	Partial	Partial	Yes	Small	Appearance of new tasks	Time-based (Converging)
[61]	2022	All	No reassignment	No	Small	Change number of UAVs	Time-based (Converging)
[62]	2020	All	No reassignment	No	Average	Appearance of new tasks + loss of UAVs	Time-based (Completion)

4.2.6. Other Extensions of Intelligent Optimization Algorithms

Other literature papers have been extending other Intelligent-based algorithms. Ghassemi et al. [63] introduced BiG-MRTA (bi-graph multi-robot task allocation) for disaster response, designed to maximize task completion rates. BiG-MRTA is tailored to address scenarios with dynamically emerging tasks that carry deadlines, considering robots' unique range and payload limitations. Simulations with up to 100 UAVs and 1000 tasks showed that BiG-MRTA performed similarly to ILP in task completion but was over 1000 times faster, outperforming the random-walk baseline in both task completion and computational efficiency, and demonstrating excellent scalability in dynamic scenarios.

Miao et al. [64] proposed a distributed immune multi-agent algorithm (DIMAA) for UAV task allocation in search and rescue missions, aiming to minimize UAV flight range. DIMAA uses an immune network framework where each UAV acts as an antibody, which is a feasible solution to the target function, interacting only within its local grid to reduce communication overhead. The agent with the optimal fitness will replace those with the worst fitness in each UAV in a single iteration cycle, improving the quality of the group and accelerating the overall evolution. Simulations with 5 UAVs, 31 tasks, and 4 new tasks demonstrated fast convergence and strong adaptability, handling dynamic task reallocation efficiently with minimal communication, ensuring task completion with robust performance.

Shao et al. [65] proposed a hybrid layered artificial potential fields (HL-APF) algorithm, combining a distributed artificial potential field with a centralized layered scheduling mechanism for cooperative UAV search in dynamic environments. The approach optimizes local searches with segmented gravitational fields while a centralized server ensures global mission planning. Simulations with 10 UAVs, 20 tasks, and 3 UAV failures showed that HL-APF increased target discovery and mission area coverage, outperforming HAPF-ACO and IAPF, while maintaining robust performance under dynamic conditions.

Table 11 compares the characteristics of the various other extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. Optimization-based algorithms, such as GA, PSO, GWO, ABC, and deep learning models, provide effective solutions for dynamic task allocation in multi-UAV systems by offering adaptability, scalability, and fast convergence. Their randomness enhances search coverage and performance in noisy environments, but their computational complexity increases significantly with problem scale, and they offer limited robustness in handling uncertainties [82]. While these algorithms perform well in real-time scenarios, producing quick solutions, they may yield suboptimal outcomes with poor scalability and flexibility compared to market-based approaches. Decentralized methods help reduce communication overhead but may depend heavily on local exchanges, leading to potential delays in disrupted networks. Additionally, many optimization algorithms prioritize fast execution over global optimization, which can impact long-term performance. Proper tuning and customization are essential to align these algorithms with specific operational needs and resource constraints, ensuring a balance between efficiency and robustness [55].

Table 11. Comparison of solution characteristics for multi-UAV dynamic task assignment extending other intelligent optimization algorithms.

Ref	Year	Paradigm	Participating UAVs	Type of Reassignment	Communication Topology	Scalability	Robustness Against Dynamic Events	Responsiveness
[63]	2022	Distributed	All	Partial	No	Large	Appearance of new tasks	Time-based (Computation)
[64]	2017	Distributed	All	All	Yes	Average	Appearance of new tasks + loss of UAVs	Time-based (Converging)
[65]	2021	Centralized-Distributed	None (Nearest one)	No reassignment	No	Average	Appearance of new tasks + loss of UAVs	Time Not Considered

4.3. Clustering-Based Algorithms

Clustering-based algorithms divide nearby or similar tasks into groups and then assign the groups to UAVs as a cluster assignment, rather than a single task assignment [23]. K-means is one of the most common clustering algorithms that build clusters based on the distances between tasks, and it is widely deployed with the task assignment problem [22]. For the multi-UAV dynamic task assignment problem with the clustering algorithm, Biswas et al. [66] introduced a nearest neighbor search (NNS) model using K-means clustering to divide the tasks into clusters equal to the number of UAVs; then, each cluster of tasks was assigned to the nearest UAV for minimizing path length and mission time. The model adapts to uncertainties like UAV failures by reallocating tasks among remaining UAVs, ensuring mission continuity. In simulations with 3 UAVs and 30 tasks, the NNS model demonstrated high task completion rates and effective task reallocation in case of UAV breakdowns, showcasing robustness in dynamic conditions.

Chen et al. [67] proposed a dynamic grouping allocation method to enhance consensus-based auction algorithms (like CBBA and IP) for maximizing task completion, minimizing waiting times, and reducing total travel costs. First, the cluster-first strategy clusters tasks into subsets equal to the number of available UAVs. Each UAV deals with tasks in its cluster until it has no task to be added, and then it communicates with other clusters for more tasks. Second, a task inclusion method is used whereby UAVs select a task from a subset of tasks (cluster) to reduce conflicts between UAVs and address the issue of becoming stuck in a local optimum. The third is a block-information-sharing method, which limits communication between UAVs so that they only share information with their group members. Simulations

with 16 UAVs and up to 32 tasks showed that the proposed approach achieved higher task assignments and shorter waiting times, with runtime scaling steadily as problem size increased, outperforming benchmarks.

Tang et al. [68] presented a multi-UAV cooperative task reallocation algorithm combining fuzzy C-means clustering with ant colony optimization (ACO) to maximize task execution rate and minimize flight distance. It details the processes of complete redistribution, partial adjustment, and group redistribution within the algorithm based on the types of emergencies. Simulations with 5 UAVs and 40 tasks showed that the group-based reallocation strategy reduced flight range by 12% and achieved a 97% task completion rate, outperforming baseline and centralized ACOs, demonstrating effective handling of emergent tasks and mission adaptability.

Table 12 compares the characteristics of the various other extensions, highlighting differences in control paradigms, reassignment strategies, and communication topologies. It focuses on the scalability of the datasets used, the robustness of the algorithms against dynamic events, and the responsiveness of the solutions in terms of time-based metrics, providing insights into their suitability for dynamic, real-world applications. Clustering-based algorithms offer several advantages and challenges for dynamic task allocation in multi-UAV systems. By grouping similar or nearby tasks, these algorithms reduce computational complexity, minimize UAV travel distances, and improve task allocation efficiency, as seen in the work by Biswas et al. [66] and Tang et al. [68]. They are particularly effective in handling uncertainties, such as UAV failures, through task-switching strategies that maintain mission continuity. Clustering methods, such as K-means and Fuzzy C-means, streamline operations by reducing the frequency of individual task assignments, which also helps manage workload distribution. Hybrid approaches, like those by Chen et al. [67], further enhance task completion and reduce waiting times by integrating clustering with auction-based strategies. However, a key challenge lies in determining the optimal cluster size, as task distribution impacts both performance and resource utilization. Additionally, while clustering algorithms perform well in dynamic environments, their performance is affected by the need for communication between clusters, which can introduce delays. Overall, clustering-based algorithms effectively balance computational efficiency with adaptability but require careful tuning to optimize cluster size and ensure scalability in complex, dynamic scenarios.

Table 12. Comparison of solution characteristics for multi-UAV dynamic task assignment extending the clustering-based algorithms. (Paradigm: distributed; scalability: average).

Ref	Year	Participating UAVs	Type of Reassignment	Communication Topology	Robustness	Responsiveness
[66]	2021	None	Partial	No	Loss of UAVs	Time-based (Completion)
[67]	2019	Partial	No reassignment	Yes	Appearance of new tasks + loss of UAVs	Time-based (Waiting)
[68]	2022	Partial	Full, Partial, or Group Redistribution (based on event type)	Yes	Appearance of new tasks + loss of UAVs + change original tasks	Time-based (Waiting)

5. Discussion

In this section, we provide a comprehensive analysis of the literature on multi-UAV dynamic task assignment, examining various dimensions to understand the current state of research and emerging trends. The analysis encompasses multiple aspects, including the distribution of publications over the years, classification of studies by their methodological approaches, controlling paradigms, and communication topologies used within multi-UAV systems. Additionally, the robustness of different algorithms in handling various dynamic events, such as the appearance of new tasks and UAV failures, is discussed. The studies are

further analyzed based on the number of participating UAVs and the scale of tasks released during reassignment processes. We also explore dataset scales, focusing on the number of tasks and UAVs involved, to understand the scalability of different approaches. This multifaceted analysis provides a holistic view of the research landscape, identifying gaps and opportunities for future work.

5.1. Publication over Years

This review spans studies published between 2013 and 2024, showcasing the growing interest in multi-UAV dynamic task assignment. As shown in Figure 3, the early years (2013–2016) witnessed limited activity, with only one or two publications annually. In 2017, research output began to increase steadily, with four studies marking the year as a turning point in the field. This growth continued through 2018 and 2019, each with three to four publications, reflecting an expanding acknowledgment of the domain's significance. A notable surge occurred in 2022, which saw an unprecedented 17 publications across various categories, underscoring the urgency to address complex, real-world challenges through innovative multi-UAV solutions. This increase can be attributed to advancements in UAV technology, increased funding opportunities, and a broader recognition of unmanned systems' strategic importance. However, 2023 experienced a decline with five publications, followed by a single publication in 2024, reflecting a slight tapering in research activity. Despite these fluctuations, the consistent overall trend highlights the sustained commitment to advancing methodologies and applications in multi-UAV dynamic task assignment, emphasizing its critical role in addressing evolving technological and operational demands.

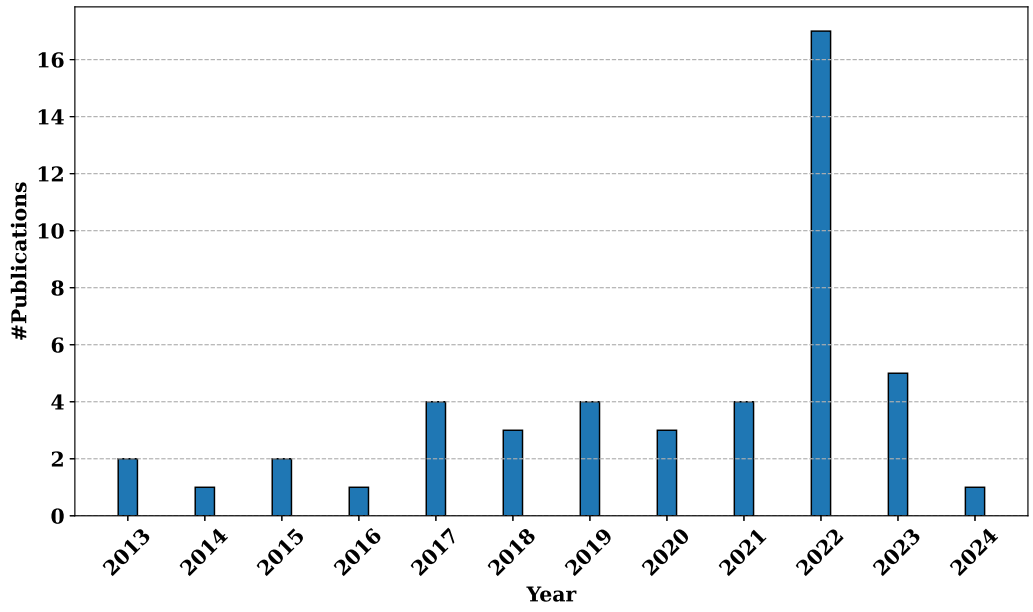


Figure 3. Total number of research publications on multi-UAV dynamic task assignment from 2013 to 2024.

5.2. Publication over Classification

Regarding the classification of the covered papers, Figure 4 provides a detailed visualization of the distribution of multi-UAV dynamic task assignment research publications from 2013 to 2024, categorized by the class of algorithms used. The analysis reveals several key insights:

- Market-based algorithms: This class has had a consistent presence throughout the years, emphasizing its enduring relevance in UAV research. Notably, 2022 witnessed a significant surge, with seven publications marking a peak in this category, suggesting

heightened interest or breakthroughs in the field. While there are no recorded publications for 2024, the steady activity over the years highlights ongoing research efforts in this area.

- Intelligent optimization algorithms: Research in this category first appeared in 2014 and gained traction in 2017. A dramatic increase occurred in 2022, with nine publications, reflecting the growing complexity of UAV operations and the corresponding demand for advanced optimization techniques. This trend continues into 2024, albeit with only one publication, indicating sustained, though reduced, interest following the peak.
- Clustering algorithms: This category exhibits a more sporadic presence, with publications recorded in 2019, 2021, and 2022, amounting to a total of three publications. The infrequent activity suggests that clustering algorithms are an emerging niche, likely driven by specific challenges or new applications in the field. No publications have been noted for 2024, potentially indicating a temporary shift in focus or the completion of foundational studies in this area.

These patterns underscore the dynamic nature of multi-UAV research, with varying levels of emphasis across algorithmic classes responding to the evolving technological and operational challenges in the field.

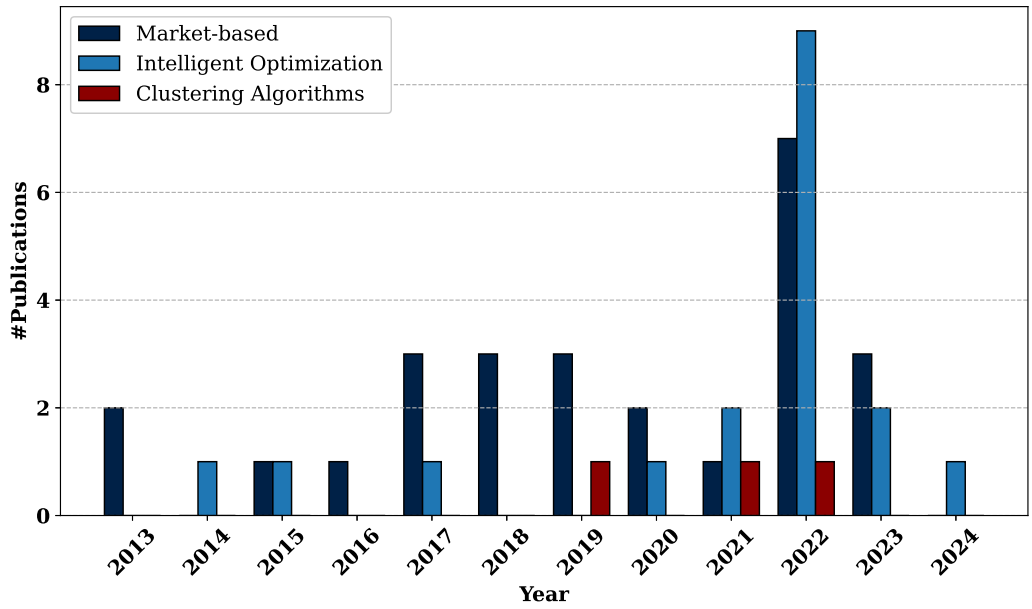


Figure 4. Year-wise distribution of research publications on multi-UAV dynamic task assignment, categorized by algorithm class from 2013 to 2024.

5.3. Controlling Paradigm

Task assignment approaches can be classified into two main paradigms based on team organization and coordination between UAVs: centralized and distributed approaches [3,83].

- Centralized approaches in UAV task allocation involve a single controller managing tasks and UAV communication. This method is efficient and reduces costs and redundant efforts. However, it faces challenges with scalability due to communication overhead, the risk of system failure, and slower response to changes, making it ideal for smaller, static environments.
- Distributed approaches, in contrast, do not rely on a central unit. Instead, UAVs communicate directly with each other, which enhances robustness and scalability. While this reduces communication demands, it requires consensus for conflict-free

task allocation, which can be slow and may yield less-than-optimal solutions. Despite these challenges, distributed approaches excel in large-scale environments.

Additionally, there is growing interest in hybrid approaches that integrate the strengths of both centralized and distributed structures—achieving "global centralization with local autonomy". This hybrid approach avoids the redundancy and heavy central load of fully centralized systems while mitigating the high capability requirements, communication complexity, and command conflicts inherent in fully distributed systems.

The distribution of controlling paradigms in multi-UAV dynamic task assignment research is visually represented in Figure 5, providing a clear overview of the distribution and focus within the field. The analysis, shown in Figure 5, reveals that the distributed paradigm dominates the field, accounting for 83% (39 papers) of the surveyed literature. This significant majority underscores the paradigm's popularity, likely due to its scalability and robustness in managing multiple UAVs in dynamic environments. On the other hand, the hybrid (centralized-distributed) approach accounts for 17% (8 papers). This lesser, yet notable proportion, indicates an emerging interest in approaches that combine features of both centralized and distributed systems, potentially aiming to harness the strengths of both in multi-UAV coordination. The rarity of the fully centralized paradigm in the surveyed literature aligns with the inherent disadvantages of centralized systems, particularly in real-time and dynamic environments.

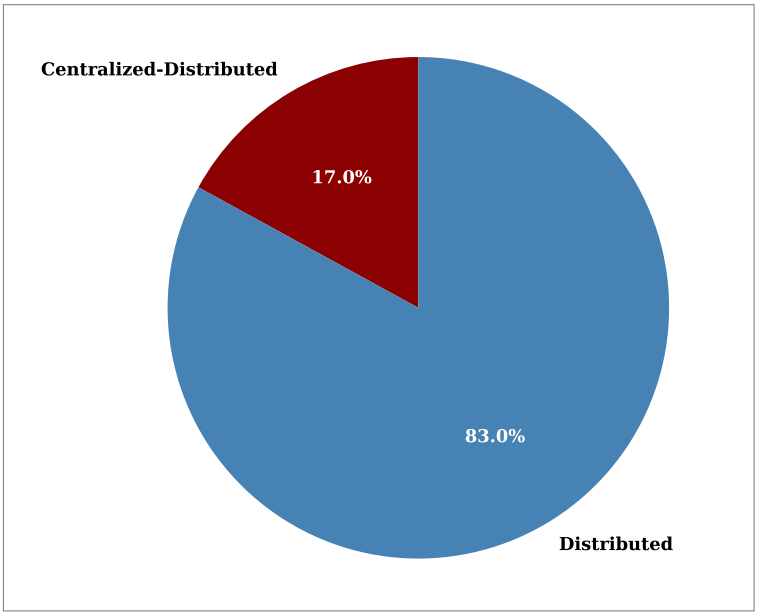


Figure 5. Distribution of controlling paradigms in multi-UAV dynamic task assignment research (2013–2024).

5.4. Communication Topology

Communication topology in multi-UAV systems is the framework that defines how UAVs communicate and interact with each other during dynamic task assignment missions. The chosen topology directly impacts the efficiency of dynamic task assignments, particularly in scenarios where quick adaptation to environmental changes and effective coordination are crucial. Figure 6 visualizes the distribution of communication topology considerations in the covered studies. The analysis reveals that a significant number of studies, represented by 66% (31 papers), do not emphasize or require a detailed description of the communication topology. This may suggest that in many cases, the specific communication structure is either assumed to be standardized or not critical to the study's objectives. In contrast, a smaller yet notable proportion of studies, accounting for 34% (16 papers),

specifically address communication topology. These studies likely focus on aspects such as network design, data-sharing protocols, and real-time communication among UAVs, reflecting the varied importance and role of communication topology in different UAV task assignment scenarios.

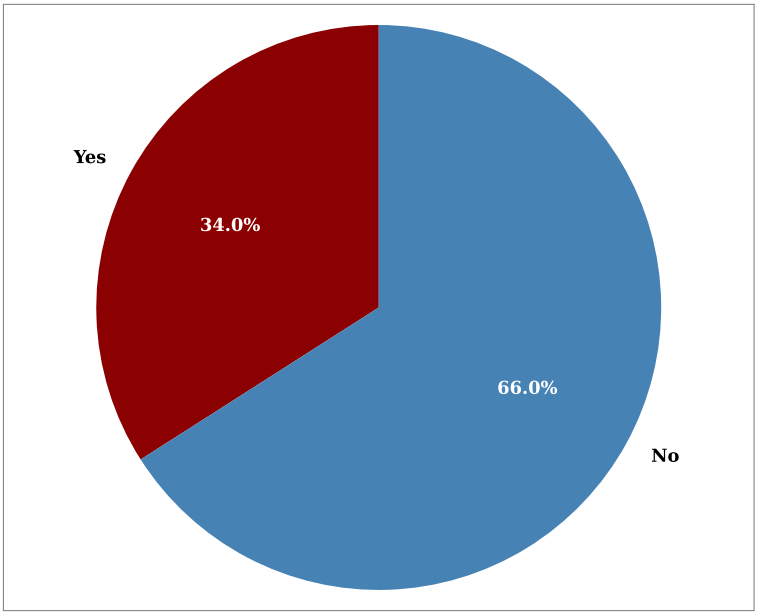


Figure 6. Distribution of communication topology considerations in multi-UAV dynamic task assignment research (2013–2024).

5.5. Robustness in Handling Dynamic Events

The ability to effectively respond to dynamic events is a cornerstone of robust multi-UAV task assignment systems. Dynamic environments, characterized by unexpected changes and evolving mission requirements, demand adaptive and resilient solutions to ensure operational success. This section delves into key aspects of robustness, offering a detailed analysis of how different approaches address the challenges posed by dynamic events. The following subsections explore the types of dynamic events these systems are designed to handle, strategies for reassigning original tasks, and the level of UAV participation in response to such events, providing a comprehensive understanding of the methodologies employed in the surveyed literature.

5.5.1. Analysis of Dynamic Event Types

In dynamic environments, the ability of multi-UAV task assignment systems to handle various unexpected events is crucial for mission success. The robustness of these systems is often tested by their response to different types of dynamic events, such as the appearance of new tasks, changes in original tasks, or the loss of UAVs. The following analysis categorizes the types of dynamic events that can impact task assignments and UAV operations, based on the reviewed literature:

1. Appearance of new tasks: This is the most common dynamic event, where new tasks emerge during a mission, requiring immediate attention. A large portion of the surveyed studies focus on scenarios where the appearance of new tasks is a critical factor [27,29–31,33,35,37,42,44,48,49,53,55,60,63]. This emphasizes the need for algorithms that can quickly adapt and reallocate resources in response to new task demands.

2. Appearance of new tasks + loss of UAVs: Several studies have investigated situations where not only new tasks appear, but UAVs are also lost, either due to technical failures or environmental factors. This combination of events requires robust strategies for reassigning tasks among the remaining UAVs, ensuring mission continuity despite the reduction in available resources [6,18,38,45,46,50,51,62,64,65,67].
3. Change of original tasks: In some cases, the objectives or requirements of originally assigned tasks change, e.g., task locations, time window, or their durations, as documented in several studies [32,34,41,47]. These scenarios demand flexibility in task management systems to accommodate alterations without compromising the overall mission.
4. Appearance of new tasks + change of original tasks: A combination of new tasks emerging and changes to existing tasks presents a complex challenge, as discussed in several studies [7,40]. The algorithms must be versatile, and capable of reassessing priorities and reallocating UAVs to meet the new task configurations effectively.
5. Appearance of new tasks + change of original tasks + loss of UAVs: This combination represents a highly dynamic and challenging scenario, highlighted in several studies [39,43,54,68]. Handling such events requires a multi-faceted approach that balances task reallocation, resource management, and system resilience.
6. Appearance of new tasks + loss of existing tasks: Some studies [28] focus on the appearance of new tasks coupled with the loss of existing ones, which could occur due to task expiration or environmental changes that render some tasks obsolete. Efficiently reallocating resources under these conditions is critical for maintaining operational efficiency.
7. Appearance of new tasks + change of original tasks + adding new UAVs + loss of UAVs: Ref. [19], which addressed this complex set of events, illustrates the need for comprehensive strategies that can manage multiple changes simultaneously, including scaling the UAV fleet up or down as needed.
8. Loss of UAVs: The isolated event of UAV loss was examined in Ref. [66], where the focus was on the system's ability to reallocate tasks among the remaining UAVs to prevent mission failure.
9. Network disconnections: A unique challenge posed by network disconnections was discussed in Ref. [36]; the authors emphasized the importance of maintaining robust communication links among UAVs to ensure smooth task coordination and execution.
10. Change in the number of UAVs: Some algorithms address the dynamic event where the number of UAVs changes during a mission, requiring flexible task assignment strategies to adapt to fluctuating resource availability [59,61].
11. No specific dynamic events mentioned: A few studies have not focused on specific types of dynamic events, potentially indicating either an assumption of a static environment or broad applicability of the proposed methods regardless of specific dynamic challenges [52,56–58].

This analysis highlights the diverse range of dynamic events that multi-UAV systems must be equipped to handle. The ability to adapt to these events is crucial for ensuring the robustness and effectiveness of UAV task assignment algorithms, particularly in complex and unpredictable environments.

5.5.2. Analysis of Original Tasks Reassignment Strategies

In multi-UAV task assignment, the ability to adapt to new dynamic events is crucial for maintaining mission effectiveness. Depending on the nature of the detected event, reassignment of tasks originally assigned to UAVs may be necessary to ensure optimal performance in handling the detected events. The strategies for handling these reassignments

vary across the literature, ranging from no reassignment to partial or full reassignment of original tasks. The reviewed literature is categorized based on their approach to task reassignment after the detection of new dynamic events, providing insight into the different methodologies adopted to maintain mission effectiveness, which is visualized in Figure 7.

1. No Reassignment: As depicted in the pie chart (Figure 7), the majority of studies (60%) choose not to reassign original tasks after detecting new dynamic events. This approach, represented by a significant portion of the reviewed literature, assumes that the original task assignments remain valid or optimal despite changes in the environment. The decision to refrain from reassignment could be based on the need to minimize disruption and maintain continuity in the mission. However, this approach may not be suitable for highly dynamic environments where task requirements or resource availability can change rapidly [18,28,32,34–38,40,41,44–48,50–53,55–58,61,62,65,67].
2. Partial reassignment: A considerable portion of studies (32%) implement partial reassignment of tasks. Refs. [6,7,19,27,29,30,33,38,39,42,43,49,60,63,66] adopted flexible approaches, selectively reallocating certain tasks while maintaining the stability of others. Partial reassignment strategies are often employed to balance responsiveness with mission stability, ensuring that critical changes are addressed without completely overhauling the task distribution.
3. Full reassignment: A smaller subset of studies (9%) advocates for the complete reassignment of all tasks upon detecting new dynamic events. This strategy, utilized in Refs. [31,64], is typically reserved for environments where the changes are significant enough to warrant a complete redistribution of tasks to ensure optimal performance. While full reassignment can be resource-intensive and may introduce delays as UAVs are reassigned, it offers the highest level of adaptability in unpredictable scenarios.

Figure 7 clearly illustrates the distribution of these strategies across the surveyed studies. The choice of reassignment strategy reflects the diverse approaches to managing dynamic events in multi-UAV task assignment, influenced by the specific goals of the mission, the nature of the environment, and the computational resources available.

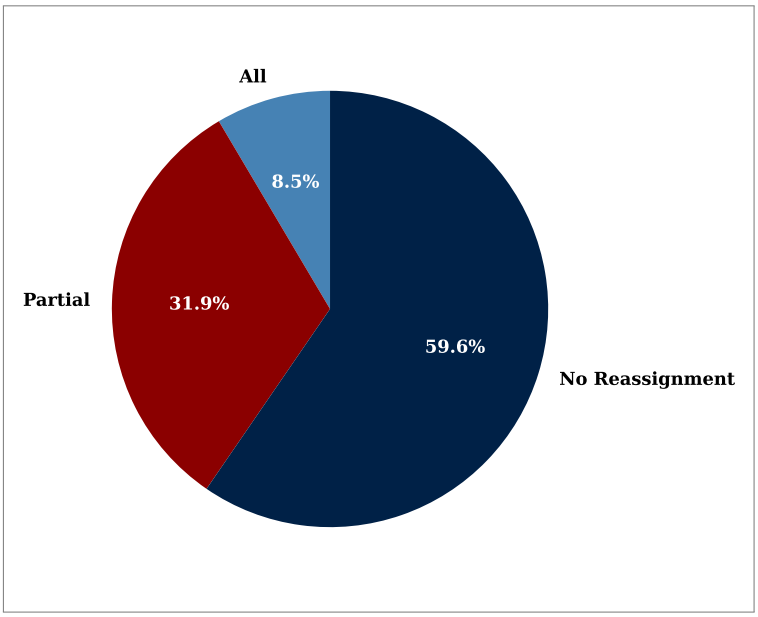


Figure 7. Distribution of task reassignment strategies in multi-UAV dynamic task assignment research.

5.5.3. Analysis of Participating UAVs in Response to Dynamic Events

In multi-UAV task assignment systems, the response to newly detected dynamic events can involve different numbers of UAVs depending on the strategy employed. The extent of participation—whether all UAVs, only a subset, or none beyond the detecting UAV—can significantly impact the system's flexibility, efficiency, and robustness.

1. All UAVs participate: The majority of studies (62%) advocate for the involvement of all available UAVs in handling detected dynamic events. This approach is represented in Refs. [6,19,28,29,31–45,48,50,51,57,58,61–64]. Utilizing the entire fleet of UAVs ensures maximum coverage and resource allocation, allowing the system to effectively manage complex or large-scale tasks that arise suddenly. However, this strategy can be resource-intensive and may lead to higher communication overhead.
2. Partial UAV participation: A significant portion of the studies (28%) implement a strategy where only a subset of UAVs is involved in responding to the new events. Refs. [7,18,27,30,46,47,49,53,55,60,67] opted for more measured responses, leveraging only the necessary number of UAVs to address the dynamic event while minimizing resource usage and communication load. This approach is often employed in scenarios where not all UAVs are required to achieve the desired outcome, thereby optimizing operational efficiency.
3. No additional UAV participation: A smaller subset of studies (11%) indicates that only the detecting UAV handles the dynamic event, with no additional UAVs participating. This minimalist approach, represented by Refs. [52,56,59,65,66], is typically used in environments where the event is localized or the resources of the detecting UAV are sufficient to manage the task independently. While this strategy reduces complexity and communication needs, it may not be suitable for more challenging or widespread events.

Figure 8 illustrates the distribution of these strategies, highlighting the preference for involving all UAVs in most scenarios. The participation level choice reflects the varying operational requirements and objectives of different UAV task assignment systems, with considerations for resource management, task complexity, and communication efficiency playing crucial roles in the decision-making process.

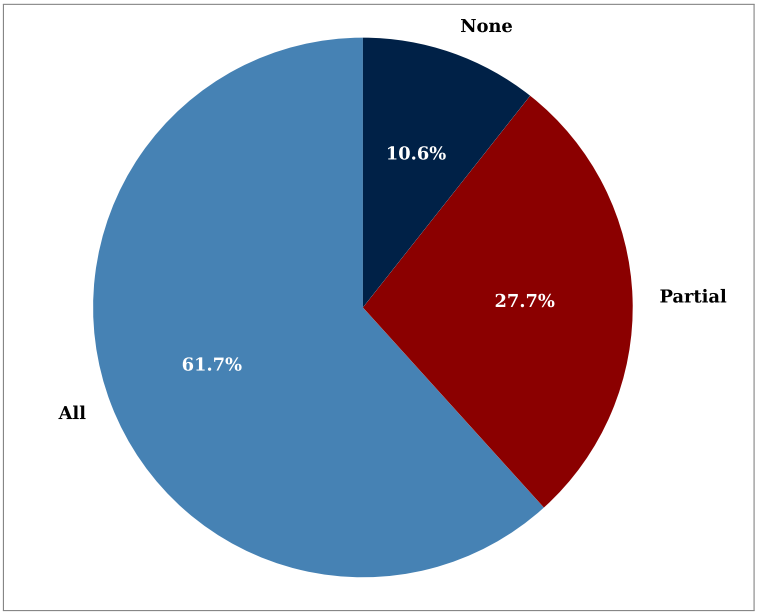


Figure 8. Distribution of UAV participation strategies in response to dynamic events in multi-UAV task assignment research.

5.6. Dataset Scales

This section presents a detailed analysis of the dataset sizes utilized in various experiments, as depicted by the number of UAVs and tasks employed. The scatter plots in Figures 9 and 10 provide a comprehensive overview of the UAVs and tasks used in multi-UAV dynamic task assignment research. Figure 10 illustrates the number of UAVs and suggests a predominant focus on small- to medium-sized UAV groups. This trend reflects a research emphasis on manageable yet realistic swarm sizes, which are optimal for testing algorithm efficiency and coordination under controlled conditions. Figure 9 depicts the number of tasks and predominantly features datasets with moderate task counts. This suggests a preference for a level of complexity that is sufficient to challenge the algorithms while remaining within practical limits for analysis and implementation.

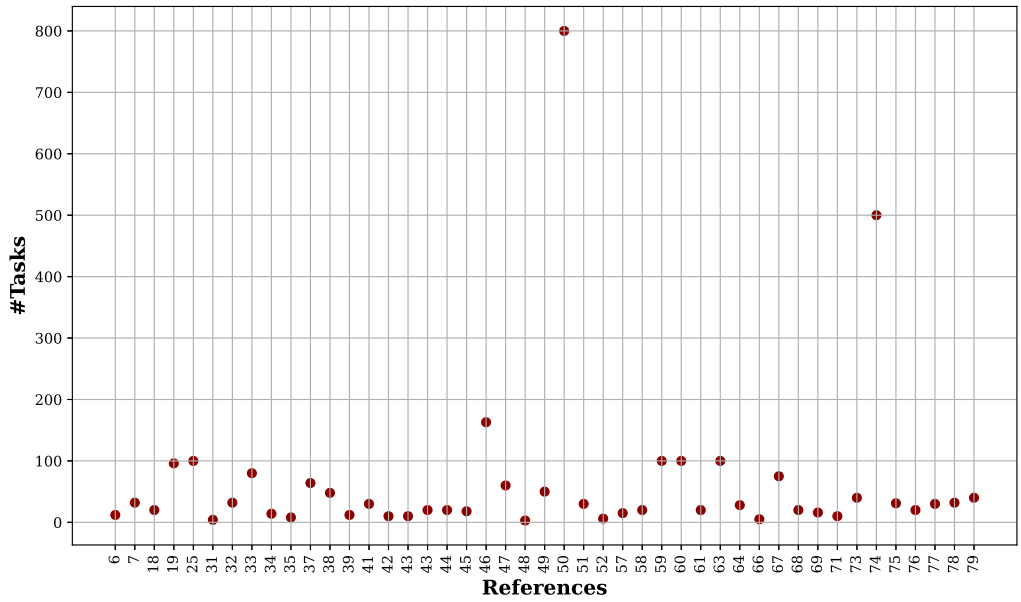


Figure 9. Scatter plot of the number of original tasks.

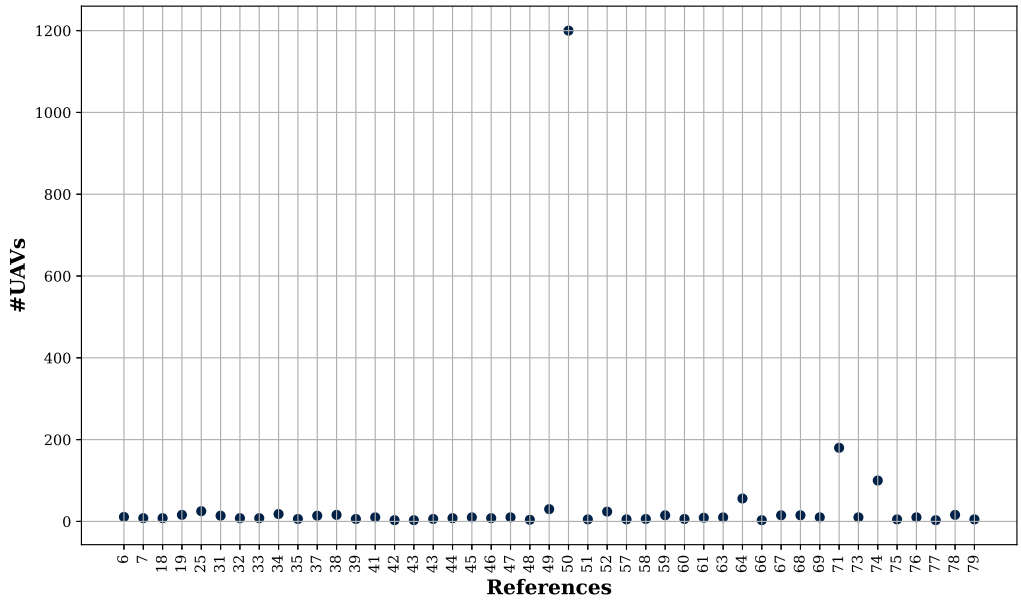


Figure 10. Scatter plot of the number of UAVs.

The classification table, shown in Table 13, further categorizes the datasets into three distinct classes: small (1–10 tasks), average (11–100 tasks), and large (100+ tasks). The studies within the small class tend to focus on precision and optimization in less complex scenarios, making them ideal for fine-tuning UAV performance and task allocation strategies. The average class, which includes the majority of the studies, represents a balanced approach that tests algorithms in moderately challenging environments, potentially reflecting a wider range of real-world applications. Although less common, studies in the large class are crucial for assessing the scalability and robustness of algorithms in high-demand, complex operations.

Table 13. Instance levels and references.

Instance Levels	References
Small (1–10)	[28,32,37,38,43,47,56,60]
Average (10–100)	[6,7,18,19,27,29–31,33–36,38–40,42,44,46,49–53,55,57,58,62,64,65]
Large (100 and above)	[41,45,63]

6. Open Issues and Research Challenges

The field of multi-UAV dynamic task assignment presents several unresolved challenges, limiting its full potential in real-world applications. Below, we discuss these challenges and propose future directions for addressing them.

- Scalability: Existing algorithms often struggle to maintain performance as the number of UAVs and tasks increases. The computational and communication demands of managing large-scale UAV fleets can significantly restrict scalability. To address this, developing lightweight and efficient algorithms that adapt to larger systems without compromising task allocation quality or real-time responsiveness is critical.
- Robustness in dynamic environments: Dynamic environments introduce frequent changes, such as the sudden appearance of tasks, UAV failures, and fluctuating mission priorities. Many current solutions lack the robustness needed to handle such unpredictability effectively. In response to this issue, enhancing algorithmic robustness through fault-tolerant mechanisms and adaptive task reassignment strategies is necessary to ensure mission continuity.
- Communication constraints: Distributed multi-UAV systems often face communication delays, bandwidth limitations, and intermittent connectivity, which degrade coordination and task allocation performance. To mitigate these challenges, designing communication-efficient algorithms that minimize bandwidth usage and ensure reliable data exchange, even in constrained or adversarial environments, is essential.
- Multi-objective optimization: UAV task assignment involves competing objectives, such as maximizing task throughput, minimizing energy consumption, and reducing completion time. Achieving a balance between these objectives is complex and often overlooked. To overcome these difficulties, multi-objective optimization frameworks that consider trade-offs between performance metrics are essential for real-world mission success.
- Hybrid approaches: While centralized approaches offer global optimization and distributed methods excel in adaptability, combining these paradigms into hybrid solutions remains underexplored. Addressing this gap, developing hybrid task assignment strategies that leverage the strengths of both centralized and distributed systems could enhance scalability and robustness.
- Real-time adaptability: Algorithms must respond to real-time changes in the environment, UAV availability, and mission parameters. Many existing solutions lack the

adaptability needed for dynamic reallocation in real-time. As a solution, designing algorithms with rapid decision-making capabilities and low computational overhead is crucial for real-time operations.

- Energy management: UAVs are constrained by limited energy resources, which affect their operational range and task execution capabilities. Task assignment solutions often neglect energy efficiency in dynamic contexts. To tackle this issue, incorporating energy-aware task allocation mechanisms that optimize energy usage while maintaining mission effectiveness is a critical area of research.
- Ethical and security concerns: The dual-use potential of UAV systems and their susceptibility to cyber-attacks pose significant ethical and security risks. To ensure responsible usage, establishing secure communication protocols, ethical guidelines, and transparent deployment practices can mitigate these risks and foster responsible usage.
- Benchmarking and standardization: The lack of standardized benchmarks and evaluation metrics makes it difficult to compare algorithms and assess their performance consistently. To foster innovation, creating realistic benchmarks that reflect diverse dynamic conditions would enable fair comparisons and drive innovation.

Addressing these open issues and challenges will require multidisciplinary research that combines advancements in algorithms, communication technologies, and ethical practices. By tackling these gaps, future work can contribute to the development of resilient and efficient multi-UAV dynamic task assignment systems that are robust, scalable, and adaptable to real-world dynamic environments.

7. Conclusions

This article has reviewed the literature on multi-UAV dynamic task assignment, analyzing a range of solutions designed to optimize mission success in dynamic environments. It underscores the critical importance of dynamic task assignment in enabling effective UAV operations, particularly in scenarios where environmental conditions, task requirements, and available resources are subject to unpredictable changes. The analysis highlights that market-based solutions are among the most widely utilized strategies due to their adaptability and scalability in distributed systems. Optimization-based algorithms and clustering solutions also play significant roles in addressing the complexities of dynamic task assignment, each offering unique advantages and limitations.

A notable trend observed in this study is the growing emphasis on enhancing existing solutions to better address the specific challenges of dynamic environments, such as rapid task reallocation, UAV failures, and energy efficiency. The comparative analysis reveals that while many studies prioritize distributed approaches, there is increasing interest in hybrid methods that integrate centralized and distributed strategies to capitalize on the strengths of both paradigms. Furthermore, the review highlights areas that remain underexplored, including communication constraints, UAV-specific limitations, and the need for multi-objective optimization.

This article not only provides a detailed overview of current approaches but also offers a foundation for advancing the field by identifying critical gaps and areas for improvement. A dedicated section on open issues and research challenges further outlines key directions for future work, emphasizing the need for more scalable, robust, and adaptive task assignment strategies. Addressing these issues will pave the way for more effective solutions capable of handling the complexities of real-world dynamic environments.

By building on these insights, future research can drive the development of innovative algorithms and methodologies that ensure resilient and efficient multi-UAV task assignment systems. This progress will enable UAV operations to achieve superior performance and reliability in increasingly complex and unpredictable scenarios.

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DURC Statement: This research focuses exclusively on the academic study and development of algorithms for multi-UAV dynamic task assignment in dynamic environments. The primary aim is to optimize mission success in scenarios such as disaster response, environmental monitoring, and search and rescue operations. The research does not pose a threat to public health or national security. The authors acknowledge the dual-use potential of UAV-related research and confirm that all necessary precautions have been taken to prevent misuse. The work adheres to relevant national and international laws and regulations regarding dual-use research. The authors advocate for ethical deployment, regulatory compliance, and responsible use of UAV technologies to foster societal benefits while mitigating risks.

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